

# CLAT Express - July 2026: Study Notes

Chapters, topics and sub-topics with key facts, memory aids and mind-map outlines.

## International Relations and News

### US-Saudi 123 Civil Nuclear Agreement

#### The US-Saudi 123 Agreement

The US and Saudi Arabia signed a Section 123 Civil Nuclear Cooperation Agreement with a bilateral safeguards arrangement, for 30 years. Section 123 of the US Atomic Energy Act, 1954 makes such agreements a legally binding prerequisite for significant US nuclear cooperation. The deal is reportedly linked to Riyadh joining the Abraham Accords and normalising relations with Israel.

- Signed for 30 years by US Energy Secretary Chris Wright and Saudi Energy Minister Prince Abdulaziz bin Salman.
- Named after Section 123 of the US Atomic Energy Act, 1954.
- A 123 Agreement is a legally binding prerequisite for significant US nuclear cooperation with another country.
- Cooperation may cover reactors, nuclear materials, components, equipment, technology and technical expertise.
- Bilateral Safeguards Arrangement ensures civilian nuclear materials are not diverted towards weapons.
- Saudi Arabia may undertake uranium enrichment in facilities developed with US support, subject to safeguards.
- The agreement does not require Saudi Arabia to adopt the IAEA Additional Protocol.
- The agreement is subject to a review period of 90 legislative days by the US Congress.
- The deal is reportedly linked to Riyadh joining the Abraham Accords and normalising relations with Israel.

Mind-map outline

123 Agreement

- Legal basis
  - Section 123, US Atomic Energy Act, 1954
  - Legally binding prerequisite
- Scope
  - Reactors, nuclear materials, components
  - Technology and technical expertise
- Safeguards
  - Bilateral Safeguards Arrangement
  - No mandatory Additional Protocol
  - Enrichment possible under safeguards
- Approval
  - Signed for 30 years
  - 90 legislative days Congress review

#### Saudi Arabia's Nuclear Vision

Saudi Arabia's nuclear programme is built on four drivers: energy diversification, large-scale desalination, meeting cooling demand and strategic technology development. The Kingdom signed the NPT in 1988 and launched its National Atomic Energy Project in 2017. The new deal drops the earlier Israel normalisation condition and allows flexibility on domestic enrichment.

- Nuclear power lets Saudi Arabia reduce its heavy dependence on crude oil and natural gas for electricity.
- The nuclear programme provides the massive energy input needed for large-scale water desalination.
- Nuclear-generated electricity provides a steady supply for air-conditioning and cooling systems.
- A nuclear programme grants access to advanced technology and reinforces Saudi status as a regional energy power.
- Saudi Arabia signed the Nuclear Non-Proliferation Treaty in 1988.

- Saudi Arabia launched its National Atomic Energy Project in 2017.
- The earlier Gold Standard required no domestic uranium enrichment or reprocessing, and normalised ties with Israel.
- The new deal drops the Israel normalisation condition and allows flexibility for US-supported domestic enrichment.

Mnemonic - Four energy drivers: Every Desert Cools Strongly

E = Energy diversification: less dependence on crude oil and natural gas for electricity; D =

Desalination: large-scale water desalination needs a massive energy input; C = Cooling: steady power for air-conditioning and cooling systems; S = Strategic technology: access to advanced technology and regional energy power status

Mind-map outline

Saudi nuclear vision

- Four drivers
  - Energy diversification
  - Large-scale desalination
  - Meeting cooling demand
  - Strategic technology development
- History
  - NPT signed 1988
  - National Atomic Energy Project 2017
- Policy shift
  - Gold Standard: no enrichment or reprocessing
  - New Deal drops Israel normalisation condition

## Nuclear Weapons History and Treaties

Nuclear weapons were developed in the US-led Manhattan Project, with the Trinity test in July 1945 and bombs dropped on Hiroshima and Nagasaki in August 1945. Treaties then tried to limit testing and spread, including the PTBT (1963), NPT (1968), CTBT (1996) and TPNW (2017). The IAEA (1957) and the NSG (1974-75) were created to manage peaceful use and export controls.

- The Manhattan Project was a secret US-led programme with the UK and Canada; Gen. Leslie Groves led engineering and J. Robert Oppenheimer led science.
- The Trinity Test in July 1945 was the first-ever nuclear explosion, conducted in the New Mexico desert.
- Little Boy (uranium gun-type) was dropped on Hiroshima on 6 August 1945; Fat Man (plutonium implosion-type) was dropped on Nagasaki on 9 August 1945.
- The Partial Test Ban Treaty (1963) banned tests in the atmosphere, outer space and underwater, but not underground.
- The Nuclear Non-Proliferation Treaty (1968) was signed to prevent the spread of nuclear weapons.
- The IAEA, established in 1957, promotes peaceful uses of atomic energy and oversees safeguards.
- The Nuclear Suppliers Group (1974-75) was formed to control the export of nuclear materials and technology.
- The CTBT (1996) was adopted by the UN to ban all nuclear explosions and has not yet entered into force.
- The TPNW (2017) entered into force in 2021 and is the first legally binding treaty to comprehensively ban nuclear weapons; nuclear powers have not joined.
- North Korea withdrew from the NPT in 2003 and later conducted nuclear tests.

Mnemonic - Scope of the Partial Test Ban Treaty: Sky, Space, Sea, but Not Underground

Sky = Atmosphere: tests banned in the atmosphere; Space = Outer space: tests banned in outer space; Sea = Underwater: tests banned underwater; Not Underground = The PTBT (1963) does not ban underground tests

Mind-map outline

Nuclear weapons history and treaties

- Manhattan Project era
  - Trinity test, July 1945
  - Hiroshima and Nagasaki, August 1945
- Treaties
  - PTBT 1963: no atmospheric or underwater tests
  - NPT 1968: stop spread of weapons

- CTBT 1996: ban all explosions, not in force
- TPNW 2017: in force 2021, nuclear powers outside
- Agencies
  - IAEA 1957: peaceful use and safeguards
  - NSG 1974-75: export control

## India's Atomic Energy Journey

India's programme was organised through the Atomic Energy Commission (1948) and the Department of Atomic Energy (1954), with Dr Homi J. Bhabha leading the AEC. Key milestones include the 1974 Pokhran-I test and the 1998 Pokhran-II tests. The 2008 Indo-US 123 Agreement ended India's nuclear isolation and was a landmark precedent.

- The Atomic Energy Commission was set up in 1948, with Dr Homi J. Bhabha as its chairman.
- The Department of Atomic Energy was created in 1954 under the Prime Minister's Office.
- The Atomic Energy Act, 1962 is India's primary law for nuclear energy.
- Tarapur Atomic Power Station, India's first commercial nuclear plant, was commissioned in 1969 under Atoms for Peace.
- Pokhran-I (1974), the Smiling Buddha test, was India's first peaceful nuclear explosion.
- Dhruva, India's largest research reactor, was commissioned at BARC in 1985.
- Pokhran-II (Operation Shakti) in 1998 led to India being declared a nuclear-weapon state.
- India's Nuclear Doctrine, adopted in 2003, follows No First Use and Credible Minimum Deterrence.
- The Indo-US Civil Nuclear Agreement (123 Agreement), signed in 2008, ended India's nuclear isolation.
- India is a Non-NPT nuclear weapons state, which makes the 2008 agreement a landmark precedent.

Mind-map outline

India's atomic energy journey

- Institutions
  - AEC set up 1948, Bhabha as chairman
  - DAE created 1954 under PMO
  - Dhruva research reactor, BARC, 1985
- Milestones
  - Pokhran-I 1974: first peaceful explosion
  - Pokhran-II 1998: declared weapons state
  - Nuclear Doctrine 2003: no first use
- Global ties
  - Tarapur 1969: first commercial plant
  - Indo-US 123 Agreement 2008

## India-Indonesia Relations

### PM's State Visit and Key Outcomes

The Prime Minister of India concluded a State Visit to Indonesia, reinforcing the India-Indonesia Comprehensive Strategic Partnership. The visit brought the Prime Minister an Indonesian honour and several agreements covering defence, industry, health, disaster management and agriculture.

- Bintang Republik Indonesia Adipurna is Indonesia's highest civilian honour.
- The Prime Minister was conferred the 35th Bintang Republik Indonesia Adipurna.
- The Prime Minister is only the 2nd Indian PM to receive it, after Jawaharlal Nehru.
- Indonesia signed a contract worth around USD 200 million for 2 batteries of BrahMos.
- Indonesia placed its 1st international order for the indigenous Astra Beyond missile.
- SAIL agreed to establish a Stainless Steel Slab Manufacturing Facility in Indonesia.
- CDSCO and its Indonesian counterpart agreed to strengthen medical product regulation.
- NDMA signed an MoU with Indonesia's Disaster Management Agency on preparedness and response.
- India announced supply of 100 tonnes of high-quality wheat seeds (DWR-162).

Mind-map outline

PM State Visit to Indonesia

- Honour
  - Bintang Republik Indonesia Adipurna

- 35th recipient
- 2nd Indian PM after Nehru
- Defence
  - BrahMos: USD 200 million
  - Astra Beyond: 1st international order
- Industry
  - SAIL stainless steel slab facility
- Health and disaster
  - CDSCO medical product regulation
  - NDMA MoU on preparedness
- Agriculture
  - 100 tonnes DWR-162 wheat seeds

### **Strategic Partnership: Defence, Maritime and Digital**

Indonesia occupies a pivotal position in India's Act East Policy, SAGAR vision and Indo-Pacific Strategy, close to the Strait of Malacca. Cooperation covers joint military exercises, maritime awareness, satellite support and digital commerce.

- Indonesia is India's 2nd largest trading partner within ASEAN.
- Indonesia is the largest economy in ASEAN.
- Major military exercises include IND-INDO CORPAT, Garuda Shakti, Samudra Shakti, Milan and Komodo.
- A 3rd India-Indonesia Security Dialogue has been proposed to strengthen defence consultations.
- India and Indonesia agreed to jointly develop Sabang Port, near the entrance to the Strait of Malacca.
- Malacca is one of the world's busiest maritime chokepoints.
- Indonesia will post a Liaison Officer at IFC-IOR to improve Maritime Domain Awareness.
- ISRO and Indonesia's BRIN jointly operate the Biak Telemetry, Tracking and Command Facility.
- Indonesia Open Network was inspired by India's Open Network for Digital Commerce model.

Mind-map outline

Strategic Partnership

- Strategic frameworks
  - Act East Policy
  - SAGAR vision
  - Indo-Pacific Strategy
  - Malacca chokepoint nearby
- Defence and exercises
  - IND-INDO CORPAT
  - Garuda Shakti, Samudra Shakti
  - Milan, Komodo
- Maritime and digital
  - Sabang Port
  - IFC-IOR liaison officer
  - Indonesia Open Network
- Trade and economy
  - 2nd largest trading partner in ASEAN
  - Largest economy in ASEAN

### **Prambanan Temple and Shared Heritage**

The Prambanan Temple Complex in Yogyakarta, locally known as Roro Jonggrang, is Indonesia's largest Hindu temple complex and India is supporting its conservation. Shared heritage and cultural diplomacy are reflected in the Tagore-Dewantara Year and a planned IIM Bangalore branch campus.

- Prambanan is locally known as Roro Jonggrang.
- Prambanan is the largest Hindu temple complex in Indonesia.
- Prambanan is the 2nd largest in Southeast Asia, after Angkor Wat in Cambodia.
- It was built in the 9th and 10th centuries CE under the Sanjaya Dynasty (Mataram Kingdom).
- It is dedicated to the Hindu Trinity of Shiva, Vishnu and Brahma.
- British surveyor Colin Mackenzie formally rediscovered the ruins in 1811.

- India will provide technical support for conservation of the Prambanan Temple Complex.
- 2026-27 is declared the Tagore-Dewantara Year of India-Indonesia Cultural and Educational Diplomacy.
- The Tagore-Dewantara Year commemorates the centenary of Rabindranath Tagore's visit to Indonesia.
- Approval was announced for an IIM Bangalore branch campus at Singhasari Special Economic Zone, East Java.
- India presented a replica of the Nalanda Copper Plate to strengthen shared civilisational links.

Mnemonic - Prambanan Trinity: Sing Very Beautifully

S = Shiva; V = Vishnu; B = Brahma

Mind-map outline

Prambanan Temple Complex

- Identity
  - Locally known as Roro Jonggrang
  - Largest Hindu temple complex in Indonesia
  - 2nd largest in Southeast Asia
- Age and dynasty
  - Built 9th-10th century CE
  - Sanjaya Dynasty, Mataram Kingdom
- Deities
  - Shiva, Vishnu, Brahma
- History and support
  - Colin Mackenzie rediscovery, 1811
  - India supports conservation

### Non-Aligned Movement (NAM)

NAM is a group of states not formally aligned with or against any major power bloc, founded in 1961 in Belgrade, Serbia. It is rooted in the Ten Principles of the 1955 Bandung Conference, and India and Indonesia were both founding members.

- NAM is a group of states not formally aligned with or against any major power bloc.
- NAM was founded in 1961 in Belgrade, Serbia.
- NAM is based on the Ten Principles of the Bandung Conference, 1955.
- Founding fathers include Jawaharlal Nehru (India), Sukarno (Indonesia), Josip Broz Tito (Yugoslavia) and Kwame Nkrumah (Ghana).
- NAM emphasised the 4Ds: Detente, Disarmament, Decolonisation and Development of the Third World.
- NAM contributed to the 1963 Partial Test Ban Treaty through declarations on nuclear testing and non-proliferation.
- NAM helped create several nuclear-weapon-free zones.
- NAM advocated for the Global South and supported anti-apartheid in South Africa and Namibia.
- NAM has supported India's candidature at the UNSC.
- NAM failed to halt the arms race and to resolve regional conflicts such as Iran-Pakistan and Russia-Ukraine.

Mnemonic - Four Ds of NAM: Dear Diplomats Drive Development

Dear = Detente; Diplomats = Disarmament; Drive = Decolonisation; Development = Development of the Third World

Mind-map outline

Non-Aligned Movement (NAM)

- Origins
  - Founded 1961, Belgrade
  - Ten Principles of Bandung, 1955
- Founders
  - Nehru (India)
  - Sukarno (Indonesia)
  - Tito (Yugoslavia)
  - Nkrumah (Ghana)
- Goals
  - 4Ds: Detente, Disarmament

- Decolonisation, Development
- Achievements
  - 1963 Partial Test Ban Treaty contribution
  - Nuclear-weapon-free zones
- Limits
  - Could not halt arms race
  - Regional conflicts unresolved

## National News and Issues

### 130th Constitution Amendment Bill

#### Three Bills and Their Scope

The 130th Constitution Amendment Bill, 2025 is one of three Bills tabled by the Home Minister, and together the Bills propose the removal of the Prime Minister, Chief Ministers and Ministers under certain conditions. A Joint Parliamentary Committee is examining the Bill and is expected to finalise and adopt its report. The other two Bills are the Government of Union Territories (Amendment) Bill, 2025 and the Jammu and Kashmir Reorganisation (Amendment) Bill, 2025.

- Introduced in 2025 as the 130th Constitution Amendment Bill
- Three Bills tabled by the Home Minister
- Bill 1: Constitution (One Hundred and Thirtieth Amendment) Bill, 2025
- Bill 2: Government of Union Territories (Amendment) Bill, 2025
- Bill 3: Jammu and Kashmir Reorganisation (Amendment) Bill, 2025
- The Bills propose removal of Prime Minister, Chief Ministers and Ministers under certain conditions
- Joint Parliamentary Committee (JPC) is examining the 130th Amendment Bill
- The JPC is expected to finalise and adopt its report

Mind-map outline

Three Bills tabled by Home Minister

- Scope of removal
  - Prime Minister
  - Chief Ministers
  - Ministers under certain conditions
- Bill titles
  - Constitution (130th Amendment) Bill
  - Government of UTs (Amendment) Bill
  - J&K Reorganisation (Amendment) Bill
- Parliamentary stage
  - Introduced in 2025
  - JPC examining the Bill
  - Report expected to be adopted

#### Removal Conditions and Authorities

The Bill's trigger is arrest and detention for 30 consecutive days on an offence punishable with five years or more. The removal authority depends on the office: the President for Union Ministers, the Governor for State Ministers and Chief Ministers, and the Lieutenant Governor for Delhi and Jammu & Kashmir. If no advice is given for a Union Minister, the Minister automatically ceases to hold office on the 31st day.

- Trigger: arrested and detained for 30 consecutive days
- The offence must be punishable with imprisonment of five years or more
- Union Ministers: Article 75 is amended
- Union Ministers: the President removes on the advice of the Prime Minister
- If no advice is given, the Minister automatically ceases to hold office on the 31st day
- A removed Minister can be reappointed after release from custody
- State Ministers and Chief Ministers: Article 164, removal authority is the Governor
- Delhi Ministers: Article 239AA, removal authority is the Lieutenant Governor

- Union Territory Ministers: Government of Union Territories Act, 1963, removal authority is the Administrator or Lieutenant Governor
- Jammu & Kashmir Ministers: J&K Reorganisation Act, 2019, removal authority is the Lieutenant Governor

Mnemonic - Thirty then Thirty-one: 30 detained, 31 gone

30 = Consecutive days of arrest and detention that trigger removal of a Union Minister; 31 = The day on which a Minister automatically ceases to hold office if no advice is given

Mind-map outline

Removal Conditions and Authorities

- Trigger
  - Arrested and detained 30 consecutive days
  - Offence punishable with 5+ years
- Removal authority
  - Union Ministers: President
  - State Ministers and CMs: Governor
  - Delhi and J&K: Lieutenant Governor
- After removal
  - No advice: ceases office on 31st day
  - Reappointment after release

## Governance Implications and Concerns

The Bill is presented as strengthening ethical governance by stopping leaders from holding power while facing serious criminal charges. It is also described as bridging a gap in the Representation of the People Act, which disqualifies legislators only after conviction. The booklet records several concerns, including misuse of arrests against opposition-led states, a threat to federalism and the presumption of innocence.

- The Bill upholds constitutional morality and ethical governance
- It stops leaders from retaining power while facing serious criminal charges
- It removes the 'governance from jail' anomaly
- It protects public trust by acting against criminalisation of politics
- It bridges a legal gap: the RPA disqualifies only post-conviction, while the Bill addresses interim arrests
- Concern: the Centre could destabilise opposition-led states through targeted arrests
- Concern: the Bill reduces the autonomy of elected state leaders, threatening federalism
- Concern: removal based solely on detention contravenes 'innocent until proven guilty'
- Concern: ED and CBI could be exploited to arrest leaders on weak charges

Mind-map outline

Governance Implications and Concerns

- Governance gains
  - Constitutional morality
  - Ends 'governance from jail'
  - Public trust in institutions
- Concerns raised
  - Political weaponisation
  - Federalism and state autonomy
  - Presumption of innocence
- Agencies at risk
  - ED and CBI misuse

## Representation of People Act and Court Rulings

Section 8 of the Representation of the People Act, 1951 sets different disqualification rules for convictions, depending on the offence and the sentence. Section 8(4), which protected sitting members who appeal within three months, was struck down by the Supreme Court. Supreme Court rulings on ministers with criminal backgrounds and on who can add grounds of disqualification form the wider legal background.

- Section 8(1): conviction for promoting enmity, rape, terrorist acts or insulting the national flag means six years' disqualification on mere conviction

- Section 8(2): conviction for hoarding or dowry offences with a sentence of at least six months; the bar continues six years after release
- Section 8(3): conviction for other offences with a sentence of not less than two years; disqualification till six years after release
- Section 8(4): sitting MPs and MLAs can continue if they appeal within three months, until the term ends or the appeal is decided, whichever is earlier
- Lily Thomas v. Union of India: the Supreme Court struck down Section 8(4) as invalid
- Manoj Narula v. Union of India (2014): no constitutional bar exists on appointing ministers with a criminal background
- Public Interest Foundation PIL (2018): only Parliament can legislate on disqualification; courts cannot add new grounds
- V Senthil Balaji case (2025): a Tamil Nadu minister reappointed post-bail had to choose between freedom and office, and he stepped down while bail continued
- Law Commission 170th Report: recommended disqualification at the framing of charges for offences punishable with five years or more

Mnemonic - Section 8 durations: Six Mere, Two Years, Three Months

Six Mere = Section 8(1): six years' disqualification on mere conviction; Two Years = Section 8(3): a sentence of not less than two years for other offences; Three Months = Section 8(4): a sitting member who appeals within three months can continue

Mind-map outline

Legal Background

- Section 8 of RPA, 1951
  - 8(1): six years on mere conviction
  - 8(2): hoarding, dowry offences
  - 8(3): other offences, two years
  - 8(4): sitting members, appeal in 3 months
- Supreme Court rulings
  - Lily Thomas: 8(4) struck down
  - Manoj Narula: no bar on ministers
  - PIL 2018: only Parliament legislates
  - Senthil Balaji: Tamil Nadu minister

## Passage of a Constitution Amendment Bill

A Constitution amendment Bill can be introduced in either House, but it must be passed by both Lok Sabha and Rajya Sabha. Once passed, it is presented to the President, who is bound to give consent, and the Bill then becomes an Act. The booklet also links the process to the Anti-Defection Law and related constitutional articles.

- The Bill can be introduced either in Lok Sabha or Rajya Sabha
- The Bill must be passed by both Lok Sabha and Rajya Sabha
- Lok Sabha strength is 545 members
- In a tie, the Presiding Officer of a house can cast his or her vote
- The passed Bill is presented to the President
- The President is bound to give consent
- After assent the Bill becomes an Act
- Related provisions: Articles 102 and 191, the Anti-Defection Law and the Lily Thomas case

Mind-map outline

Passage of a Constitution Amendment Bill

- Introduction
  - Either Lok Sabha or Rajya Sabha
- Passage
  - Must be passed by both Houses
  - Tie: Presiding Officer may vote
- Final stage
  - Presented to the President
  - President bound to give consent
  - Bill becomes an Act
- Related provisions

- Articles 102 and 191
- Anti-Defection Law

## Polity & Governance

### Autonomous Hill Development Councils expanded across Ladakh

#### Why the AHDC expansion matters

The Ladakh Administration has approved Autonomous Hill Development Councils (AHDCs) in all 7 districts after an in-principle agreement with the Ministry of Home Affairs. The move follows Ladakh becoming a Union Territory without a Legislative Assembly in 2019, which makes decentralised local institutions central to governance. Before the expansion, elected councils existed only in Leh and Kargil.

- Ladakh Administration approved AHDCs in all 7 districts
- In-principle agreement with the Ministry of Home Affairs
- Agreement aims to extend Article 371 constitutional safeguards to the UT
- Ladakh became a UT without a Legislative Assembly in 2019
- Decentralised local institutions are central to governance in Ladakh
- Earlier, elected councils existed only in Leh and Kargil

Mind-map outline

Why AHDCs are expanding

- All 7 districts approved
- MHA in-principle agreement
- UT without Legislative Assembly (2019)
- Earlier councils only in Leh and Kargil

#### Legal basis, powers and finances of AHDCs

AHDCs function under the Ladakh Autonomous Hill Development Council Act and are created through legislation, applying only in Ladakh. Section 3(1) empowers the Government to establish a Council in every district through an official notification. Their powers are mainly administrative and developmental, covering land allotment, district cadre recruitment and promotions, and district development planning. Each AHDC maintains a Council Fund and can levy taxes, fees and other lawful charges for local governance.

- Functions under the Ladakh Autonomous Hill Development Council Act
- Created through legislation and applicable only in Ladakh
- Section 3(1): Government can set up a Council in every district by official notification
- Power over land allotment
- Power over district cadre recruitment and promotions
- Power over district development planning
- Sectors include health, education, tourism, infrastructure and social welfare
- Each AHDC maintains a Council Fund
- Can levy taxes, fees and other lawful charges for local governance
- No judicial authority

Mnemonic - Sectors under AHDC powers: Hungry Elephants Tickle Interesting Sheep

H = Health; E = Education; T = Tourism; I = Infrastructure; S = Social welfare

Mind-map outline

AHDC legal basis, powers, finance

- Legal basis
  - Ladakh AHDC Act
  - Section 3(1): notification per district
  - Applies only in Ladakh
- Powers
  - Land allotment
  - Cadre recruitment and promotions
  - District development planning

- Health, education, tourism, infrastructure, welfare
- Finance
  - Council Fund
  - Taxes, fees, lawful charges
- Limit
  - No judicial authority

### **AHDCs versus 6th Schedule councils and Article 371**

AHDCs and 6th Schedule Autonomous District Councils are both local bodies, but they differ in how they are created, where they apply and what powers they hold. AHDCs are created through legislation and apply only in Ladakh, while 6th Schedule councils have constitutional status under Articles 244(2) and 275(1). Article 371 and Articles 371A to 371J, under Part XXI, provide special safeguards for indigenous culture, customary laws, land rights and local employment.

- AHDCs: created through legislation and apply only in Ladakh
- 6th Schedule councils: constitutional status under Articles 244(2) and 275(1)
- 6th Schedule councils apply in notified tribal areas of Assam, Meghalaya, Mizoram and Tripura
- AHDCs: mainly administrative and developmental powers, no judicial authority
- 6th Schedule councils: legislative, executive, financial and judicial powers
- 6th Schedule councils can establish village courts under customary tribal laws
- Article 371 and Articles 371A to 371J fall under Part XXI
- Article 371 aims to protect indigenous culture, customary laws, land rights, local employment, tribal identity and administrative autonomy

Mnemonic - AHDC versus 6th Schedule status: Act makes AHDC; Constitution makes the Sixth Act = AHDCs are created through legislation; Constitution = 6th Schedule councils have constitutional status under Articles 244(2) and 275(1)

Mind-map outline

AHDC vs 6th Schedule councils

- AHDC
  - Created through legislation
  - Only in Ladakh
  - Administrative and developmental powers
- 6th Schedule
  - Articles 244(2) and 275(1)
  - Assam, Meghalaya, Mizoram, Tripura
  - Village courts under customary laws
- Article 371
  - Part XXI
  - Protects culture, land rights, employment

### **Ladakh's profile, demands and 2025 reservation rules**

Ladakh, known as the Land of High Passes or Maryul, has been administered by the Government of India as a Union Territory since 31 October 2019. Its people demand inclusion under the 6th Schedule, restrictions on land ownership to prevent outsider influx, and a Legislative Assembly. A reservation regulation approved on 2 June 2025 and effective from 3 June 2025 raises the total reservation cap to 85 per cent, with EWS quotas excluded.

- Known as the Land of High Passes, or Maryul
- Administered by the Government of India as a UT since 31 October 2019
- Capital: Leh, with Kargil as the winter capital
- Demand: inclusion under the 6th Schedule for constitutional protection
- Demand: land ownership restrictions to prevent outsider influx
- Demand: a Legislative Assembly for representative governance
- Reservation (Amendment) Regulation, 2025 approved on 2 June 2025
- Effective from 3 June 2025; affects government jobs and educational admissions
- Total reservation cap raised to 85 per cent, EWS quotas excluded

Mnemonic - 2025 reservation regulation: Two approved, Three effective, Eighty-five capped  
Two = Regulation approved by the President on 2 June 2025; Three = Effective from 3 June 2025;  
Eighty-five = Total reservation cap of 85 per cent, EWS quotas excluded

Mind-map outline

Ladakh profile, demands, reservation

- Profile
  - Land of High Passes (Maryul)
  - UT since 31 Oct 2019
  - Winter capital: Kargil
- Demands
  - 6th Schedule inclusion
  - Land ownership restrictions
  - Legislative Assembly
- Reservation 2025
  - Approved 2 June 2025
  - Effective 3 June 2025
  - Total cap 85 per cent

## Delhi Electric Vehicle Policy 2.0

### Policy Overview and Targets

Delhi EV Policy 2.0 was notified by the Government of NCT of Delhi and is scheduled to come into force from 1 July 2026, remaining effective until 31 March 2030. It builds upon the achievements of Delhi EV Policy 2020 and aims to turn Delhi into a leading electric mobility hub. The policy targets at least 30% electrification of Delhi's total vehicle fleet by 31 March 2030, with a focus on high-usage and high-emission transport segments.

- Notified by the Government of NCT of Delhi
- In force from 1 July 2026
- Effective until 31 March 2030
- Builds upon the achievements of Delhi EV Policy 2020
- Electrification target: at least 30% of Delhi's total vehicle fleet by 31 March 2030
- Focus: electrification of high-usage and high-emission transport segments
- Priority segments: 2-wheelers, 3-wheelers, commercial goods carriers, school buses, app-based aggregator fleets, government-owned transport fleets
- Objective: widespread adoption of zero-emission vehicles
- Objective: reduce urban air pollution and greenhouse gas emissions

Mnemonic - Key objectives recall: Zebras Pick Charging Stations

Z = Zero-emission vehicle adoption; P = Pollution (urban air) and greenhouse gas reduction; C = Charging and battery-swapping infrastructure; S = Sustainable public and commercial transport

Mind-map outline

Policy overview and targets

- Key objectives
  - Zero-emission vehicle adoption
  - Reduce urban air pollution
  - Reduce greenhouse gas emissions
  - Charging and battery-swapping infrastructure
  - Sustainable public and commercial transport
- Electrification target
  - At least 30% of total fleet
  - By 31 March 2030
  - Focus: high-usage, high-emission segments

## Registration Rules and Phase-out

New registrations in Delhi move to electric models in stages. From 1 January 2027 only electric 3-wheelers and electric N1 category light commercial trucks can be newly registered, and from 1 April 2028 new 2-wheeler registrations are restricted to electric models. Existing petrol and diesel vehicles may keep operating until the end of their prescribed service life. Schools must also electrify part of their bus fleets.

- From 1 January 2027: only electric 3-wheelers and electric N1 category light commercial trucks newly registered
- From 1 April 2028: registration of new 2-wheelers restricted exclusively to electric models
- Existing petrol and diesel vehicles registered in Delhi may operate until the end of their prescribed service life
- N1 category refers to light commercial trucks
- Schools must convert at least 10% of their bus fleet into electric buses within 2 years of policy implementation
- Long-term objective: 30% electrification of school bus fleets by 2030

Mnemonic - Registration timeline: Three Nice Trucks in 27, Two Wheels in 28

Three Nice Trucks in 27 = From 1 January 2027, only electric 3-wheelers and electric N1 light commercial trucks are newly registered; Two Wheels in 28 = From 1 April 2028, new 2-wheeler registration is restricted exclusively to electric models

Mind-map outline

Registration rules and phase-out

- Electric-only registration
  - 1 January 2027: electric 3W and N1 trucks
  - 1 April 2028: new 2W electric only
- Existing vehicles
  - Petrol and diesel run until service life ends
- School buses
  - Convert at least 10% within 2 years
  - 30% of school bus fleet by 2030

## Investment, Support and Charging

The policy plans total investment of approximately 15,000 crore. Government support is restricted exclusively to battery electric vehicles and not to hybrid vehicles. The plan includes installing over 30,000 public charging stations across Delhi. Financial support and investment is handled by the Delhi Transport Department, and infrastructure development by Delhi Transco Limited.

- Total planned investment: approximately 15,000 crore
- Government support restricted exclusively to battery electric vehicles, not hybrid vehicles
- Over 30,000 public charging stations planned across Delhi
- Financial support and investment implemented by Delhi Transport Department
- Infrastructure development implemented by Delhi Transco Limited
- Key objective: strengthen charging and battery-swapping infrastructure
- Key objective: encourage sustainable public and commercial transport

Mind-map outline

Investment, support and charging

- Investment
  - Total planned investment approx 15,000 crore
- Government support
  - Only battery electric vehicles, not hybrids
- Charging
  - Over 30,000 public charging stations
- Implementing agencies
  - Financial support: Delhi Transport Department
  - Infrastructure: Delhi Transco Limited

## National Schemes: NEMMP, FAME and PM E-DRIVE

Delhi's policy sits within national electric mobility efforts. The National Electric Mobility Mission Plan was launched in 2013, and the government committed to achieve 30% e-mobility by 2030. FAME India, launched by the Ministry of Heavy Industries & Public Enterprises, is part of that mission plan and offers subsidies for several vehicle types. PM E-DRIVE, launched in 2024 with the Ministry of Heavy Industries as nodal ministry, builds on FAME-II and the Electric Mobility Promotion Scheme 2024.

- National Electric Mobility Mission Plan launched in 2013
- Government committed to achieve 30% e-mobility by 2030
- FAME India launched by Ministry of Heavy Industries & Public Enterprises
- FAME India is a part of the National Electric Mobility Mission Plan
- FAME subsidy can be claimed for two-wheelers, three-wheelers, electric and hybrid cars and electric buses
- PM E-DRIVE launched in 2024; scheme period 2024-2026
- PM E-DRIVE nodal ministry: Ministry of Heavy Industries
- PM E-DRIVE objectives: reduce upfront EV cost and expedite EV adoption
- PM E-DRIVE objectives: establish public EV charging and promote EV manufacturing ecosystem
- PM E-DRIVE builds on FAME-II and Electric Mobility Promotion Scheme 2024

Mnemonic - National schemes sequence: Thirteen Mission, Fame Inside, Drive Twenty-Four  
Thirteen Mission = National Electric Mobility Mission Plan launched in 2013; Fame Inside = FAME India is a part of the National Electric Mobility Mission Plan; Drive Twenty-Four = PM E-DRIVE launched in 2024

Mind-map outline

National schemes

- NEMMP
  - Launched in 2013
  - Government committed to 30% e-mobility by 2030
- FAME India
  - Launched by Ministry of Heavy Industries
  - Part of the NEMMP
  - Subsidy: 2W, 3W, electric and hybrid cars, e-buses
- PM E-DRIVE
  - Launched in 2024; scheme period 2024-2026
  - Nodal ministry: Ministry of Heavy Industries
  - Reduce upfront EV cost

## PM Surya Ghar: Muft Bijli Yojana

### PM Surya Ghar: Muft Bijli Yojana scheme

PM Surya Ghar: Muft Bijli Yojana is a rooftop solar scheme launched in 2024 and described as the world's largest domestic rooftop solar initiative. Its aim is to provide free electricity up to 300 units to households by installing rooftop solar panels. The scheme offers a subsidy of up to 40% on panel installation and targets 1 crore households by 2027, with the Ministry of New and Renewable Energy as nodal ministry. A parallel Model Solar Village initiative asks each district to establish one such village to promote energy self-reliance.

- Launched in 2024
- World's largest domestic rooftop solar initiative
- Aim: free electricity up to 300 units per household
- Subsidy of up to 40% for rooftop solar panel installation
- Target: rooftop solar in 1 crore households by 2027
- Nodal ministry: Ministry of New and Renewable Energy
- World Bank approved USD 890 million to support the programme
- Model Solar Village: one per district to promote energy self-reliance
- Model Solar Village eligibility: population above 5,000 (2,000 in special category states)
- Top contributing states: Gujarat, Maharashtra, Rajasthan

Mnemonic - PM Surya Ghar numbers: Three Forty One Twenty-Seven

Three = Free electricity up to 300 units per household; Forty = Subsidy of up to 40% on panel installation; One = Target of 1 crore households; Twenty-Seven = Target year 2027

Mind-map outline

PM Surya Ghar: Muft Bijli Yojana

- Launched in 2024
- Aim: 300 free units per household
- Subsidy up to 40%
- Target: 1 crore households by 2027
- Nodal ministry: MNRE

## International Solar Alliance (ISA)

The International Solar Alliance (ISA) is a treaty-based intergovernmental organisation of solar-rich countries lying between the Tropic of Cancer and the Tropic of Capricorn. It was launched jointly by India and France in 2015 during COP21 in Paris. Its membership comprises 112 members and 15 signatory countries, and its headquarters is in Gurugram.

- Launched jointly by India and France in 2015 during COP21 (Paris)
- Treaty-based intergovernmental organisation of solar-rich countries
- Covers countries between the Tropic of Cancer and the Tropic of Capricorn
- Membership: 112 members and 15 signatory countries
- Non-tropical countries can join but are excluded from voting rights
- After the 2020 Amendment, all UN Member States are eligible to join
- UN General Assembly granted observer status to ISA in 2021
- Headquarters: Gurugram (National Institute for Solar Energy of India)
- Mission: mobilise USD 1 trillion in solar investments by 2030
- Mission: install 1,000 GW of solar capacity

Mnemonic - ISA launch: Two Nations, Fifteen, Paris

Two Nations = ISA was launched jointly by India and France; Fifteen = Launch year 2015; Paris = Launch took place during COP21 in Paris

Mind-map outline

International Solar Alliance

- Founding
  - 2015, launched by India and France
  - Announced at COP21, Paris
- Membership
  - 112 members, 15 signatory countries
  - Non-tropical countries: join, no voting
  - All UN members eligible after 2020 Amendment
- Recognition and base
  - Headquarters: Gurugram
  - UN General Assembly observer status, 2021
- Mission
  - USD 1 trillion solar investment by 2030
  - Install 1,000 GW solar capacity

## India's solar position and records

According to IRENA Renewable Energy Statistics 2026, India added 37 GW of solar capacity in 2025 and became the world's second largest solar growth market. India overtook Japan to become the world's third largest solar power producer in 2025. The country also hosts Bhadla Solar Park, India's largest solar plant at 2245 MW, and Omkareshwar, India's largest floating solar plant at 600 MW.

- India added 37 GW of solar capacity in 2025 (IRENA Renewable Energy Statistics 2026)
- India emerged as the world's 2nd largest solar growth market
- China added 315 GW of solar in 2025 and remained the largest
- Global solar capacity rose from 1880 GW in 2024 to 2392 GW in 2025
- India overtook Japan to become the world's 3rd largest solar power producer in 2025

- India ranked 3rd globally in total renewable energy installed capacity with 250.52 GW
- Top solar states: Rajasthan, Gujarat, Tamil Nadu, Karnataka, Maharashtra
- Bhadla Solar Park (Jodhpur, Rajasthan, 2245 MW): India's largest solar plant
- Omkareshwar (Khandwa, Madhya Pradesh, 600 MW): India's largest floating solar plant
- Rajarhat (Kolkata): India's first floating solar plant (2015)

Mind-map outline

India's solar position and records

- 2025 position
  - Added 37 GW solar
  - 2nd largest solar growth market
  - 3rd largest solar power producer
- Record plants
  - Bhadla Solar Park: 2245 MW
  - Omkareshwar floating: 600 MW
  - Rajarhat: first floating plant (2015)

## Global initiatives and climate commitments

India's COP26 climate commitments, called Panchamrita, include net zero by 2070 and 500 GW of non-fossil energy capacity by 2030. The Global Solar Facility, launched at COP27 in 2022, provides payment guarantees to de-risk solar investments. The OSOWOG concept, first proposed by India in 2018, advocates trans-national solar power connectivity through integrated regional grids.

- Panchamrita: India's climate commitments at COP26 in Glasgow
- Net zero target by the year 2070
- Non-fossil energy capacity of 500 GW by 2030
- Meet 50 percent of energy requirements from renewable energy by 2030
- Reduce total projected carbon emissions by one billion tonnes from now till 2030
- Global Solar Facility launched at COP27 (2022) to provide payment guarantees and de-risk solar investments
- STAR-C trains engineers, entrepreneurs and policymakers in solar technologies
- One Sun, One World, One Grid (India and UK, COP26) advocates trans-national solar connectivity
- India first proposed the OSOWOG concept in 2018 during ISA's Inaugural Assembly
- OSOWOG aims to integrate regional grids across Asia, Middle East, Africa and Europe for 24-hour renewable power

Mnemonic - Panchamrita targets: Nice Green Homes Beat Inflation

N = Net zero by 2070; G = 500 GW non-fossil energy capacity by 2030; H = 50 percent of energy from renewables by 2030; B = Cut projected carbon emissions by one billion tonnes; I = Reduce carbon intensity of the economy by 2030

Mind-map outline

Global initiatives and climate commitments

- Panchamrita (COP26, Glasgow)
  - Net zero by 2070
  - 500 GW non-fossil capacity by 2030
  - 50% energy from renewables by 2030
  - 1 billion tonne projected emission cut
- Global Solar Facility
  - Launched at COP27, 2022
  - Payment guarantees to de-risk investment
- OSOWOG
  - Proposed by India in 2018
  - 24-hour renewable power via linked grids

## Proposed NFSA Amendment for AAY Households

## Proposed AAY Entitlement Change

The Union Government has proposed amending Section 3(1) of the National Food Security Act 2013 to change foodgrain entitlement for Antyodaya Anna Yojana households. The current rule gives each AAY household a fixed 35 kg per month irrespective of family size. The proposal shifts the basis to individuals, at 7 kg per person per month, with a maximum of 35 kg per household.

- Amendment seeks to modify Section 3(1) of NFSA 2013 relating to AAY foodgrain entitlement
- Current provision: every AAY household receives 35 kg per month irrespective of family size
- Proposed provision: every individual in an AAY household receives 7 kg per month
- Proposed ceiling: maximum 35 kg per household
- The proposal moves the basis of AAY entitlement from household to individual
- The proposal has been made by the Union Government

Mnemonic - 7 per head, 35 per home: 7 per head, 35 per home

7 = kg per individual per month under the proposal; 35 = kg maximum per household per month under the proposal

Mind-map outline

Proposed AAY Entitlement Change

- Current design
  - Fixed 35 kg per household per month
  - Same amount regardless of family size
- Proposed design
  - 7 kg per individual per month
  - Maximum 35 kg per household
- Legal reference
  - Modify Section 3(1) of NFSA 2013

## Rationale and Limits of the Amendment

The Union Government argues that the household-based AAY entitlement creates inequities within the category. Small households receive more foodgrain per person than larger ones, and the amendment would make distribution proportional to household size. The government also says the change reduces disparities with Priority Household beneficiaries, but the amendment does not address inclusion errors.

- Government argues the household-based entitlement creates inequities within the AAY category
- Per capita inequality: small households receive more foodgrain per person than larger households
- Better equity: distribution becomes proportional to household size
- Rationalisation: allocation aligned more closely with nutritional requirements
- Uniformity: reduces disparities between AAY and Priority Household beneficiaries
- Limitation: the amendment does not address inclusion errors, where ineligible beneficiaries continue to receive NFSA benefits

Mind-map outline

Rationale and Limits

- Reasons given by government
  - Household-based entitlement inequitable
  - Per capita inequality
  - Proportional to household size
  - Rationalisation with nutritional requirements
  - Uniformity between AAY and PHH
- Limitation
  - Inclusion errors not addressed

## NFSA 2013 and Beneficiary Categories

The National Food Security Act 2013 was enacted to ensure food and nutritional security through subsidised foodgrains under the Targeted Public Distribution System. It made foodgrain access a legal right for two-thirds of the population. AAY households get 35 kg per month and PHH households get 5 kg per person per month at subsidised rates.

- NFSA 2013 was enacted to ensure food and nutritional security
- It provides subsidised foodgrains through the Targeted Public Distribution System

- It made foodgrain access a legal right for two-thirds of the population
- AAY is meant for the poorest of the poor households
- AAY families get 35 kg per month, fixed per household
- PHH covers other eligible households
- PHH get 5 kg per person per month
- AAY entitlement is fixed per household; PHH entitlement is per person

Mind-map outline

NFSA 2013 Framework

- Objective
  - Food and nutritional security
  - Subsidised foodgrains via Targeted PDS
- Legal right
  - Two-thirds of population
- Beneficiary categories
  - AAY: poorest of the poor households
  - PHH: other eligible households

## PDS Background and PMGKAY

The Public Distribution System was introduced by the British during World War II, after the 1942-43 Bengal famine, to control food prices and ration foodgrains. It was initially focused on urban food security and expanded to tribal and poor areas after the Green Revolution. PMGKAY (2020) added extra free foodgrains for NFSA beneficiaries.

- PDS in India was introduced by the British during World War II
- It was introduced after the 1942-43 Bengal famine to control food prices and ration foodgrains
- Initially focused on urban food security
- Expanded to tribal and poor areas after the Green Revolution in the 1970s and 1980s
- Essential goods distributed: wheat, sugar, rice and kerosene
- Administered by the Ministry of Consumer Affairs, Food and Public Distribution
- RPDS (1992) aims to extend PDS access to remote, hilly areas through an area-based approach
- PMGKAY (2020) provided an extra 5 kg of free foodgrains to AAY and PHH beneficiaries
- PMGKAY covered 81.35 crore people alongside regular PDS entitlements
- Kerala set up one of India's earliest PDS in 1962; Tamil Nadu introduced universal free rice distribution in 2011

Mnemonic - Four stages of PDS history: Famine, War, Green, Gift

Famine = PDS introduced after the 1942-43 Bengal famine; War = PDS introduced by the British during World War II; Green = PDS expanded to tribal and poor areas after the Green Revolution; Gift = PMGKAY (2020) provided extra free foodgrains

Mind-map outline

PDS Background and PMGKAY

- Origins
  - Introduced by British in WWII
  - Followed 1942-43 Bengal famine
- Expansion
  - Urban food security first
  - Tribal and poor areas after Green Revolution
- Later additions
  - RPDS: area-based approach for remote areas
  - PMGKAY 2020: extra 5 kg free foodgrains

## Economics

### India-UK Comprehensive Economic and Trade Agreement (CETA)

## Scope and Entry into Force

The India-UK CETA came into force on 15 July 2026, together with the Double Contribution Convention. The booklet calls it a next-generation free trade agreement that goes beyond conventional tariff reduction to cover market access, regulatory transparency and economic partnership. Its coverage spans goods and services, digital trade, investment, government procurement, labour, environment, innovation, SMEs and regulatory cooperation. It is also the first Indian FTA to give tariff concessions on imported passenger vehicles through tariff rate quotas.

- CETA and the Double Contribution Convention came into force on 15 July 2026.
- CETA is a next-generation FTA that goes beyond conventional tariff reduction.
- Coverage includes trade in goods and services, digital trade and investment.
- Also covers government procurement, labour, environment, innovation, SMEs and regulatory cooperation.
- Addresses market access, regulatory transparency and economic partnership.
- First Indian FTA to give tariff concessions on imported passenger vehicles through tariff rate quotas.
- The quota approach protects domestic industry while allowing phased market access.

Mind-map outline

CETA scope and timing

- Entry into force
  - 15 July 2026
  - Same date as DCC
- Next-generation FTA
  - Beyond tariff reduction
  - Market access, regulatory transparency
- Coverage areas
  - Goods, services, digital trade, investment
  - Procurement, labour, environment
  - Innovation, SMEs, regulatory cooperation
- Passenger vehicles
  - First Indian FTA with TRQ concessions
  - Protects domestic industry

## Tariff Cuts and Market Access

India plans to cut duties on 90% of tariff lines, with 85% becoming tariff-free within 10 years. Indian exports get zero-duty access for 99% of products, while auto import tariffs fall to 10% under a quota system from over 100%. Whisky and gin tariffs drop from 150% to 75% and reduce to 40% by the 10th year. The deal also adds 100 annual visas for Indian professionals, and bilateral trade of USD 56.8 billion is expected to double by 2030.

- India plans to cut duties on 90% of tariff lines.
- 85% of tariff lines become tariff-free in 10 years.
- Zero-duty access for 99% of Indian exports to the UK.
- Auto import tariffs cut to 10% under the quota system, down from over 100%.
- Whisky and gin tariffs cut from 150% to 75%, reducing to 40% by the 10th year.
- 100 additional annual visas for Indian professionals.
- Bilateral trade of USD 56.8 billion (GBP 42.6 billion), expected to double by 2030.
- UK is India's 4th largest export destination; India ranks 11th as UK's trading partner.
- Data exclusivity is excluded from the FTA to safeguard India's generic pharma sector.

Mnemonic - Tariff cut figures: Ninety cut, eighty-five free in ten

Ninety cut = India plans to cut duties on 90% of tariff lines; Eighty-five free in ten = 85% of tariff lines become tariff-free in 10 years

Mind-map outline

Tariff cuts and market access

- Tariff lines
  - 90% of lines: duties cut
  - 85% tariff-free in 10 years
- Exports to UK
  - Zero duty for 99% of Indian exports

- Autos
  - Auto tariff 10% under quota
  - Down from over 100%
- Whisky and gin
  - 150% to 75% initially
  - 40% by 10th year
- Trade and visas
  - Bilateral trade USD 56.8 billion
  - 100 additional annual visas
- Pharma safeguard
  - Data exclusivity excluded

## Double Contribution Convention

The Double Contribution Convention came into force along with CETA and is aimed at social security contributions. Eligible Indian professionals and employers are exempt from UK National Insurance contributions for up to 5 years. The exemption applies provided contributions continue in India. The purpose is to ensure professionals avoid paying social security contributions in both India and the UK.

- India and UK signed the Double Contribution Convention Agreement.
- Exempts eligible Indian professionals from paying UK National Insurance contributions.
- Exempts eligible Indian employers from paying UK National Insurance contributions.
- The exemption covers up to 5 years.
- Condition: social security contributions must continue in India.
- Purpose: professionals avoid paying social security contributions in both India and UK.
- Came into force on 15 July 2026 along with CETA.

Mind-map outline

Double Contribution Convention

- Who benefits
  - Eligible Indian professionals
  - Eligible Indian employers
- Exemption
  - UK National Insurance contributions
  - Up to 5 years
- Condition
  - Contributions continue in India
- Purpose
  - Avoid social security in both countries

## Brexit and Trade Realignment

The India-UK FTA is part of Britain's post-Brexit strategy. The UK joined the EU in 1973 and formally left it on 31 January 2020, losing EU market access in the process. The UK and other western nations are diversifying trade away from China, with India as a China-plus-one alternative. India also moved away from RCEP towards bilateral FTAs such as the UK deal.

- The UK joined the EU in 1973, when it was known as the European Economic Community.
- A referendum held in June 2016 saw around 17.4 million people opt for Brexit.
- The UK formally left the EU on 31 January 2020, ending 47 years of membership.
- A transition period under the withdrawal agreement ran until the end of 2020.
- Brexit meant loss of EU market access, so the UK sought new trade partnerships.
- UK and Western nations diversify trade away from China, with India as China-plus-one alternative.
- India exited RCEP over trade concerns and shifted focus to bilateral FTAs such as the UK deal.
- US tariffs and geopolitical tensions push nations toward stable and predictable trade relations.
- The India-UK FTA offers a stable trade framework for economic predictability and long-term growth.

Mnemonic - Brexit timeline: Join at 73, vote at 16, leave at 20

Join at 73 = UK joined the EU (then EEC) in 1973; Vote at 16 = Brexit referendum held in June 2016;

Leave at 20 = UK formally left the EU on 31 January 2020

Mind-map outline

Brexit and trade realignment

- Brexit timeline
  - Joined EU 1973
  - Referendum June 2016
  - Formal exit 31 January 2020
- Post-Brexit need
  - Loss of EU market access
  - Seek new trade partners
- China-plus-one
  - Diversify away from China
  - India as alternative
- India's realignment
  - Exited RCEP over trade concerns
  - Bilateral FTAs preferred
- Global uncertainty
  - US tariffs, geopolitical tensions
  - Stable, predictable trade sought

## **RBI Plans Trial of Polymer (Plastic) Currency Notes**

### **Polymer Banknotes: Concept and Proposal**

Polymer banknotes are printed on a specialised plastic substrate, primarily BOPP, instead of cotton-based banknote paper. RBI has revived a proposal for such notes that was first made in 2009, and BRBNMPL invited global bids for the polymer substrate. Australia was the first country to introduce a complete series of polymer banknotes, in 1988.

- Polymer banknotes use a specialised plastic substrate, primarily BOPP.
- BOPP stands for Biaxially Oriented Polypropylene.
- Polymer notes replace conventional cotton-based banknote paper.
- Australia was the first country to introduce a complete series of polymer banknotes, in 1988.
- The initial proposal for polymer notes in India dates to 2009.
- RBI revived the proposal to introduce durable polymer-based banknotes.
- RBI invited bids to procure polymer notes in the current initiative.
- BRBNMPL, RBI's note-printing subsidiary, invited global bids for a BOPP-based polymer substrate.
- BRBNMPL was established in 1995 in Bengaluru.

Mnemonic - BOPP expansion: Big Orange Pandas Pounce

B = Biaxially; O = Oriented; PP = Polypropylene

Mind-map outline

Polymer Banknotes: Concept and Proposal

- BOPP substrate, plastic
- Replaces cotton-based paper
- Australia: first complete series, 1988
- Initial proposal: 2009
- RBI revived proposal
- BRBNMPL invited substrate bids

### **Advantages of Polymer Notes**

Polymer notes are more resistant to tearing, less vulnerable to water and moisture, and more durable under frequent circulation. They can carry advanced anti-counterfeiting features that are much harder to replicate than on paper. Because individual notes last longer, fewer new notes need to be manufactured to keep supply steady.

- More resistant to tearing.
- Less vulnerable to water and moisture.
- More durable under frequent circulation.
- Capable of incorporating advanced anti-counterfeiting features.
- Polymer notes remain usable for significantly longer periods.
- Longer life means less frequent replacement cycles.

- Fewer new notes need to be manufactured and issued to keep supply steady.
- Polymer substrates allow security features that are much harder to replicate than on paper.
- Less frequent replacement lowers the long-term environmental footprint.

Mnemonic - Advantages of polymer notes: Tough Wallets Dislike Snatchers

T = Resistant to tearing; W = Less vulnerable to water and moisture; D = Durable under frequent circulation; S = Capable of advanced anti-counterfeiting features

Mind-map outline

Advantages of Polymer Notes

- Durability
  - Tear resistance
  - Water and moisture resistance
  - Frequent circulation
- Security
  - Anti-counterfeiting features
  - Harder to replicate than paper
- Lifecycle
  - Longer usable life
  - Less frequent replacement
  - Fewer new notes issued

## Costs, Supply Chain and Environmental Challenges

Polymer notes cost significantly more to produce than paper currency, and their production cost is tied to global oil prices. The notes are made from polypropylene, and reliance on imported polypropylene creates import dependency. Plastic waste needs a specialised collection and recycling system, and the transition requires changes to ATMs, vending machines and cash-handling machines.

- Polymer notes cost significantly more to produce than conventional paper currency.
- Production cost is tied to global oil prices.
- Polymer notes are made from polypropylene.
- Reliance on imported polypropylene creates import dependency.
- Unlike cotton-based paper, polymer notes raise concerns about plastic waste.
- Plastic waste needs an effective specialised collection and recycling system.
- Transition requires recalibration of ATMs, currency vending machines and automated cash-handling machines.
- Changing to polymer notes creates substantial secondary expenses for banking and retail infrastructure.

Mind-map outline

Costs, Supply Chain and Environment

- Cost
  - Costs significantly more than paper
  - Tied to global oil prices
- Supply
  - Made from polypropylene
  - Import dependency
- Environment
  - Plastic waste concern
  - Needs collection and recycling system
- Infrastructure
  - ATMs and vending machines updated
  - Banking and retail systems modified

## Currency Management and Printing in India

Currency and coinage fall under the Union List, and the Reserve Bank of India Act, 1934 provides the principal legal framework for issuing banknotes. RBI issues notes, estimates currency requirements and ensures circulation of good-quality notes. SPMCIL and BRBNMPL together operate four presses that print all banknotes in India.

- Currency, coinage, legal tender and foreign exchange fall within the Union List.

- Reserve Bank of India Act, 1934 is the principal legal framework for issuing banknotes.
- Coinage Act, 2011 deals with the design, production and minting, and circulation of coins.
- RBI roles include issuing banknotes and estimating the country's currency requirements.
- Four currency printing presses print all banknotes of India.
- SPMCIL is a Government of India enterprise operating 2 presses, including one at Nashik, Maharashtra.
- BRBNMPL is a wholly owned RBI subsidiary operating 2 presses, at Mysuru (Karnataka) and Salboni (West Bengal).
- Bank Note Paper Mill India Pvt. Ltd. manufactures banknote paper domestically.
- RBI introduced the Clean Note Policy in 1999.
- Government designs and mints coins, and RBI distributes them through its currency management network.

Mnemonic - Key legal years: RBI thirty-four, coins twenty-eleven

1934 = RBI Act: principal legal framework for issuing banknotes; 2011 = Coinage Act: design, production, minting and circulation of coins

Mind-map outline

Currency Management in India

- Legal framework
  - Union List: currency and coinage
  - RBI Act, 1934
  - Coinage Act, 2011
- RBI roles
  - Issues banknotes
  - Estimates currency needs
  - Ensures good-quality circulation
- Printing
  - SPMCIL: 2 presses
  - BRBNMPL: 2 presses
  - Bank Note Paper Mill: paper
- Government role
  - Issues Rs 1 note
  - Designs and mints coins

## Summits, Reports, Indices

### 16th India-Japan Annual Summit 2026

#### Summit Outcomes

The 16th India-Japan Annual Summit was held in New Delhi during the first official India visit of Japan's Prime Minister Sanae Takaichi, from July 1 to 3, 2026. The summit produced outcomes in defence, economic security, energy, artificial intelligence, trade and disaster risk reduction. Some outcomes launch new programmes, while others are agreements to convene further meetings or to review existing arrangements.

- 16th India-Japan Annual Summit held in New Delhi; Japan PM Sanae Takaichi's first official India visit, July 1-3, 2026
- Defence: agreement in principle on remaining technical aspects of UNICORN (Unified Complex Radio Antenna), the first Defence Co-Development Programme between the two countries
- 4th India-Japan 2+2 Ministerial Meeting to be convened later in 2026
- Joint Declaration on Economic Security Cooperation adopted
- Priority areas: semiconductors, critical minerals, ICT, pharmaceuticals, clean energy technologies
- Cooperative Biogas for Growth Initiative launched; Japan supports India's objective of 1,000 Biogas and Organic Fertiliser Plants
- AI Strategic Dialogue launched; Joint Statement on Artificial Intelligence Cooperation adopted
- Trade: accelerate review of CEPA; work towards a target of ¥10 Trillion in Japanese investment in India
- Memorandum of Cooperation signed supporting a USD 10 Billion investment package

- Japan supports India hosting the 4th UN World Conference on Disaster Risk Reduction in 2030; carries forward the Sendai Framework 2015-2030

Mnemonic - Economic Security Priority Areas: Silly Cats Invent Pills Cleverly

Silly = Semiconductors; Cats = Critical minerals; Invent = Information and Communication

Technologies; Pills = Pharmaceuticals; Cleverly = Clean energy technologies

Mind-map outline

Summit outcomes

- Defence: UNICORN project
  - Technical aspects agreed in principle
  - First Defence Co-Development Programme
- Economic security
  - Joint Declaration adopted
  - Five priority areas
- Energy and AI
  - Biogas for Growth launched
  - AI Strategic Dialogue launched
- Trade and disaster risk
  - CEPA review to be accelerated
  - UN conference support, 2030

## India-Japan Partnership Beyond the Summit

Japan's support for Subhas Chandra Bose's Azad Hind Fauj, formed to fight colonial rule, is a historical link in the relationship. Japan is India's largest official development assistance donor and its fifth largest investor. The partnership also covers joint military exercises, a lunar space mission and several global initiatives.

- Subhas Chandra Bose established Azad Hind Fauj (Indian National Army) with Japanese support to fight colonial rule
- Bilateral trade in 2025-26: \$27.48 billion
- Japan is the 5th largest investor in India (FDI)
- Japan is India's largest Official Development Assistance donor since 1958
- Joint exercises: Malabar (naval, with US and Australia), Milan (multilateral naval), JIMEX (navy), Dharma Guardian (army), SHINYUU Maitri (air force)
- Space: Lunar Polar Exploration Mission (LUPEX), also called Chandrayaan-5
- Global programmes: International Solar Alliance; Coalition for Disaster Resilient Infrastructure; LeadIT; India-Japan-Australia Supply Chain Resilience Initiative
- Frameworks listed under Skill-Up: Indo-Pacific Oceans Initiative, SAGAR, MAHASAGAR, QUAD, ASEAN, BIMSTEC, UNCLOS, G4 for UNSC Reform

Mind-map outline

India-Japan partnership

- History and trade
  - Azad Hind Fauj with Japanese support
  - Bilateral trade 2025-26: \$27.48 billion
- Aid and investment
  - Largest ODA donor since 1958
  - 5th largest investor (FDI)
- Defence and space
  - Five joint exercises
  - LUPEX (Chandrayaan-5)
- Global programmes
  - International Solar Alliance
  - Supply Chain Resilience Initiative

## Mumbai-Ahmedabad High-Speed Rail Corridor

The Mumbai-Ahmedabad High-Speed Rail is a flagship India-Japan project that uses Shinkansen technology and forms part of India's National Rail Plan 2030. The corridor runs from Mumbai in Maharashtra to Ahmedabad in Gujarat and also passes through Dadra and Nagar Haveli. Japan finances most of the cost through a JICA loan and will supply Shinkansen train sets for the line.

- Route: Mumbai (Maharashtra) to Ahmedabad (Gujarat); length 508.17 km
- Length split: 348 km in Gujarat, 156 km in Maharashtra, 4 km in Dadra & Nagar Haveli
- Stations: 8 in Gujarat and 4 in Maharashtra
- Technology: Shinkansen; part of India's National Rail Plan 2030
- Japan will provide India with 3 Shinkansen train sets, including the E10 and E3 series
- India to become the 1st country outside Japan to adopt the E10 Shinkansen high-speed train, designed by JR East (Japan)
- Funding: over 80% financed by Japan (JICA loan) at 0.1% interest, repayable in 50 years
- Implemented by National High-Speed Rail Corporation Limited
- Deadlines: Surat-Bilimora stretch by 2026; full corridor by 2028

Mnemonic - Rail Funding Terms: Eighty at a tenth, fifty years to pay

Eighty = Over 80% of the project financed by Japan through a JICA loan; a tenth = 0.1% interest rate; fifty years = Loan repayable in 50 years

Mind-map outline

Mumbai-Ahmedabad High-Speed Rail

- Route and length
  - Mumbai (Maharashtra) to Ahmedabad (Gujarat)
  - 508.17 km total
  - Gujarat 348 km; Maharashtra 156 km
  - Dadra & Nagar Haveli 4 km
- Technology and trains
  - Shinkansen technology
  - 3 Shinkansen train sets from Japan
  - E10: first outside Japan for India
- Funding
  - JICA loan over 80%
  - 0.1% interest
  - Repayable in 50 years
- Deadlines
  - Surat-Bilimora stretch by 2026
  - Full corridor by 2028

## Green Hydrogen and the National Green Hydrogen Mission

Hydrogen is the most common element in nature but exists only in combination with other elements, so it must be extracted from compounds such as water. Green hydrogen is made by splitting water through electrolysis using an electrolyser powered by renewable energy, and the booklet says it is not commercially viable at present. The National Green Hydrogen Mission, with the Ministry of New and Renewable Energy as nodal ministry, aims to decarbonise sectors and build domestic manufacturing.

- Hydrogen is the most common element in nature but exists only in combination with other elements
- Hydrogen has to be extracted from naturally occurring compounds such as water
- Green hydrogen: water is split by electrolysis using an electrolyser powered by renewable energy
- Green hydrogen is not commercially viable at present
- Nodal ministry: Ministry of New and Renewable Energy
- Objectives: decarbonise energy, industrial and mobility sectors; develop indigenous manufacturing; create export opportunities for green hydrogen and its derivatives
- SIGHT: Strategic Interventions for Green Hydrogen Transition Programme
- SHIP: Strategic Hydrogen Innovation Partnership (PPP for R&D)
- Expected by 2030: Rs 1 lakh crore in fossil fuel import savings

Mnemonic - Green Hydrogen Programmes: SIGHT the transition, SHIP the innovation  
SIGHT = Strategic Interventions for Green Hydrogen Transition Programme; SHIP = Strategic Hydrogen Innovation Partnership, a PPP for R&D

Mind-map outline

Green hydrogen

- Production
  - Electrolysis of water
  - Electrolyser powered by renewable energy
- Mission objectives
  - Decarbonise energy, industry, mobility
  - Build indigenous manufacturing
  - Export opportunities for GH2 and derivatives
- Programmes
  - SIGHT: transition programme
  - SHIP: PPP for R&D
- Status
  - Not commercially viable at present
  - Rs 1 lakh crore fossil fuel import savings

## UN Sustainable Development Report 2026

### Evolution of the SDGs

The idea of sustainable development was defined in 1987 and then built into successive global commitments. Milestones such as the Millennium Development Goals, the Johannesburg Summit and Rio+20 moved the agenda forward. The 2015 adoption of the 2030 Agenda introduced 17 SDGs supported by 169 targets.

- 1987: Brundtland Commission defined the concept of sustainable development
- 1992: Earth Summit (UNCED) held in Rio de Janeiro, Brazil
- 2000: Millennium Development Goals adopted, with 8 goals
- 2002: Johannesburg World Summit on Sustainable Development, also called Rio+10
- 2012: Rio+20 Conference laid the foundation for the SDGs
- 2015: 2030 Agenda and 17 SDGs adopted
- The 2030 Agenda was adopted by all 193 UN Member States
- 17 Integrated Global Goals are supported by 169 Specific Targets
- Objective: eliminate poverty, reduce inequality and protect the environment by 2030

Mnemonic - SDG milestones: Bold Mangoes Juice Rio Tomorrow

B = Brundtland Commission defined sustainable development in 1987; M = Millennium Development Goals adopted in 2000 with 8 goals; J = Johannesburg World Summit in 2002, also called Rio+10; R = Rio+20 Conference in 2012 laid the foundation for the SDGs; T = 2030 Agenda with 17 SDGs adopted in 2015

Mind-map outline

SDG evolution, 1987 to 2015

- 1987: Brundtland concept
- 2000: MDGs, 8 goals
- 2002: Johannesburg, Rio+10
- 2012: Rio+20 base for SDGs
- 2015: 2030 Agenda, 17 SDGs
  - 169 Specific Targets
  - Adopted by 193 UN Member States

### Earth Summit 1992 and its outcomes

The UN Conference on Environment and Development in Rio de Janeiro, popularly called the Earth Summit, aimed to integrate environmental protection with sustainable development. Its key outcomes were the Rio Declaration, Agenda 21 and the Forest Principles. The Rio Declaration sets out guiding principles, Agenda 21 is a non-binding action plan, and the Forest Principles are non-legally binding.

- Popularly known as the Earth Summit (1992)

- Objective: integrate environmental protection with sustainable development
- Rio Declaration: a set of 27 Guiding Principles
- Rio principles emphasise the precautionary approach and the polluter pays principle
- Common but Differentiated Responsibilities is one of the Rio principles
- Agenda 21: a non-binding action plan for sustainable development
- Agenda 21 covers poverty, sustainable consumption and major groups
- Forest Principles: non-legally binding principles on conservation and management of forests
- Outcomes laid the foundation for the Kyoto Protocol (1997), Johannesburg Summit (2002) and Paris Agreement (2015)

Mnemonic - Rio outcomes: Rio's Agenda Forests

R = Rio Declaration: set of 27 Guiding Principles; A = Agenda 21: non-binding action plan; F = Forest Principles: non-legally binding

Mind-map outline

Earth Summit 1992

- Rio Declaration
  - 27 Guiding Principles
  - Precautionary approach
  - Polluter pays principle
  - Common but differentiated responsibilities
- Agenda 21
  - Non-binding action plan
  - Covers poverty and sustainable consumption
- Forest Principles
  - Non-legally binding
- Laid foundations for later agreements
  - Kyoto Protocol (1997)
  - Johannesburg Summit (2002)
  - Paris Agreement (2015)

## SDR 2026 findings and India's position

The 11th Sustainable Development Report 2026, released by the UN Sustainable Development Solutions Network, ranks countries on their SDG performance using a 0-100 score. India moved up to 94th place with an overall SDG Index score of 68.3 out of 100. Nearly half of all targets show only slow or limited progress, and the report identifies Zero Hunger as India's most critical development challenge.

- 11th Sustainable Development Report 2026, released by UN SDSN
- Annual assessment of SDG progress, published since 2016
- Assesses 193 UN Member States on a 0-100 SDG score
- India ranked 94th, its highest position to date
- India's overall SDG Index score: 68.3 out of 100
- India improved from 112th position in 2015 to 94th in 2026
- Approx 33.3% of India's SDG targets are expected to be achieved by 2030
- Nearly half of all targets show only slow or limited progress
- About 15% of targets have experienced setbacks compared with their 2015 baseline
- Top performers: Finland, Sweden, Denmark; lowest: Chad, Central African Republic, South Sudan

Mind-map outline

UN SDR 2026

- Ranking
  - India 94th, highest to date
  - Top: Finland, Sweden, Denmark
  - Lowest: Chad, CAR, South Sudan
- Global progress
  - Nearly half of targets slow
  - About 15% of targets set back
- India performance
  - SDG Index 68.3
  - Up from 112th in 2015

- 33.3% of targets expected by 2030
- Zero Hunger most critical challenge
- Domestic bodies
  - NITI Aayog: nodal agency
  - Ministry of Panchayati Raj localises SDGs

## Climate finance

Climate finance refers to financing from public, private or alternative sources that supports mitigation and adaptation against climate change. It rests on the Polluter Pays principle and CBDR-RC, and multilateral climate funds are coordinated by the UNFCCC. Persistent challenges include gaps between national needs and funding, slow approvals and much lower per-capita approved funding for Least Developed Countries.

- Climate finance: local, national or transnational financing for climate action
- Sources: public, private or alternative financing
- Principles: Polluter Pays, and Common but Differentiated Responsibility and Respective Capability (CBDR-RC)
- Multilateral climate funds are coordinated by the UNFCCC
- Global Environment Facility (GEF): operating entity of the financial mechanism (1994)
- Adaptation Fund gives developing countries full ownership of adaptation projects
- Clean Development Mechanism: emission-reduction projects in developing nations
- Green Climate Fund estd. 2010 (COP 16), with LDCF and SCCF under it
- Loss and Damage Fund (2023): financial assistance to nations most affected by climate change
- Challenges: slow rate of approvals and gaps between national needs and NDC finance

Mind-map outline

Climate finance

- Principles
  - Polluter Pays
  - CBDR-RC
- Multilateral funds (UNFCCC)
  - GEF (1994)
  - Adaptation Fund
  - CDM
  - Green Climate Fund (2010)
    - LDCF and SCCF under it
- Long-term finance
  - Cancun Agreements: scaled-up funds
  - Loss and Damage Fund (2023)
- Challenges
  - Slow rate of approvals
  - Gap versus national needs

## Panchayat Advancement Index 2.0

The Panchayat Advancement Index 2.0 is a nationwide performance assessment framework for Gram Panchayats and Traditional Local Bodies. It grades local bodies from A+ to D on a composite score from 0 to 100. Tripura had the highest share of Front Runners, and no Gram Panchayat reached the Achiever grade.

- Nationwide performance assessment framework for Gram Panchayats and Traditional Local Bodies
- Recognised as the first comprehensive index globally to evaluate rural local governance
- Uses objective and verifiable indicators
- Based on a composite score from 0 to 100
- Achiever (A+) above 90; Front Runner (A) 75-89
- Performer (B) 60-74; Aspirant (C) 40-59; Beginner (D) below 40
- Translates 17 SDGs into 9 localised SDGs
- Ministry of Panchayati Raj localises SDGs through this index
- Top performers: Tripura (highest share of Front Runners), then Kerala and Odisha
- No Gram Panchayat could reach the Achiever category

Mnemonic - PAI grades: Nice Friends Play Carrom Daily

N = Achiever (A+): composite score above 90; F = Front Runner (A): composite score 75-89; P = Performer (B): composite score 60-74; C = Aspirant (C): composite score 40-59; D = Beginner (D): composite score below 40

Mind-map outline

Panchayat Advancement Index 2.0

- Scope
  - Gram Panchayats and Traditional Local Bodies
- Grades (composite 0-100)
  - Achiever A+: above 90
  - Front Runner A: 75-89
  - Performer B: 60-74
  - Aspirant C: 40-59
  - Beginner D: below 40
- Thematic excellence
  - Poverty-free and enhanced livelihoods
  - Healthy Panchayats
- Top performers
  - Tripura, then Kerala and Odisha

## Defence, Space and S&T

### Vikram-1: India's Private Orbital Rocket

#### Mission and Orbit Basics

Skyroot Aerospace of Hyderabad launched Vikram-1 on Mission Aagaman, its maiden flight, from the Satish Dhawan Space Centre at Sriharikota. The flight placed payloads into Low Earth Orbit and made India the third country, after the US and China, to demonstrate private orbital launch capability. The booklet also separates sub-orbital rockets, which do not place a satellite into orbit, from orbital rockets, which do.

- Vikram-1 is India's first privately built orbital launch vehicle.
- Maiden flight was Mission Aagaman, launched from Satish Dhawan Space Centre, Sriharikota.
- The flight placed payloads into Low Earth Orbit.
- India became the 3rd country, after US and China, to demonstrate private orbital launch capability.
- Vikram-1 is a 4-stage orbital launch vehicle.
- The mission was authorised and facilitated by IN-SPACe.
- Vikram-1 is named after Dr Vikram Sarabhai, father of India's space programme.
- A sub-orbital rocket goes to space but does not place a satellite into orbit.
- An orbital rocket places a satellite into orbit around the Earth.
- Low Earth Orbit lies between 160 km and 2,000 km above Earth's surface.

Mnemonic - Sub-orbital vs orbital: Sub stops short; Orbit goes all the way

Sub = A sub-orbital rocket goes to space but does not place a satellite into orbit; Orbit = An orbital rocket places a satellite into orbit around the Earth

Mind-map outline

Mission and Orbit Basics

- Launch
  - Mission Aagaman, maiden flight
  - Satish Dhawan Space Centre, Sriharikota
- Achievement
  - 3rd country after US and China
  - Private orbital launch capability
- Orbit types
  - Sub-orbital: no satellite placed in orbit
  - Orbital: satellite placed in orbit
  - Low Earth Orbit: 160 km to 2,000 km

## Propulsion and Structure

The first three stages of Vikram-1, called Kalam Engines, use advanced solid propellants. The upper stage uses a 3D-printed liquid-fuelled Raman Engine for precise orbital insertion and manoeuvres. The rocket extensively uses all-carbon composite structures, and the booklet also highlights Agnilet, a single-piece 3D-printed semi-cryogenic engine.

- Stages 1 to 3 of Vikram-1 are called Kalam Engines.
- The Kalam Engines use advanced solid propellants.
- The upper stage features a 3D-printed liquid-fuelled Raman Engine.
- The Raman Engine is used for precise orbital insertion and manoeuvres.
- Vikram-1 extensively uses all-carbon composite structures.
- Agnilet is the world's first single-piece 3D-printed semi-cryogenic engine.
- Agnibaan SOrTeD is the world's first rocket with a fully 3D-printed engine.
- SOrTeD stands for SubOrbital Technology Demonstrator.

Mind-map outline

Propulsion and Structure

- Kalam Engines (stages 1-3)
  - Advanced solid propellants
- Raman Engine (upper stage)
  - 3D-printed, liquid-fuelled
  - Precise orbital insertion and manoeuvres
- Structures
  - All-carbon composite structures
- Related engines
  - Agnilet: single-piece 3D-printed semi-cryogenic
  - Agnibaan SOrTeD: fully 3D-printed engine

## Payloads on Board

Vikram-1 carries scientific, technological, commercial and symbolic payloads. SCOPE, developed by Skyroot Aerospace, validates onboard spacecraft systems and mission technologies during flight. The Embrace robotic arm, created by Cosmoserve Space, is linked to future applications such as space debris removal and in-orbit servicing.

- Vikram-1 carries scientific, technological, commercial and symbolic payloads.
- SCOPE, developed by Skyroot Aerospace, validates onboard spacecraft systems and mission technologies during flight.
- The Embrace robotic arm, created by Cosmoserve Space, stays attached to the payload deck during its demonstration.
- Embrace is linked to future applications such as space debris removal and in-orbit servicing.
- SOLARAS is an indigenous CubeSat that demonstrates indigenous satellite technology in orbit.
- Cosmic Bloom is India's first lab-grown diamond jewellery sent into space, developed by Cosmos Diamonds.
- An 18K gold rocket model carries micro-sculptures of Sir C.V. Raman, Dr Vikram Sarabhai and Dr A.P.J. Abdul Kalam.
- The gold rocket model is credited to Telangana-based artist Ajay Kumar Mattewada.

Mind-map outline

Payloads on Vikram-1

- Payload types
  - Scientific, technological, commercial, symbolic
- Technology payloads
  - SCOPE: validates systems in flight
  - Embrace: robotic arm from Cosmoserve Space
  - SOLARAS: indigenous CubeSat
- Symbolic and commercial payloads
  - Cosmic Bloom: lab-grown diamond jewellery
  - 18K gold rocket model with micro-sculptures

## Policy, Regulators and FDI

Indian Space Policy 2023 opened the complete space value chain to non-government entities.

IN-SPACE works as the single-window regulator for private space activities, while NewSpace India Limited serves as ISRO's commercial arm. FDI limits are set separately for each segment of the space sector.

- Indian Space Policy 2023 opened the complete space value chain to non-government entities.
- NewSpace India Limited serves as ISRO's commercial arm and was established in 2019.
- IN-SPACE functions as a single-window regulator for private space activities.
- IN-SPACE stands for Indian National Space Promotion and Authorisation Centre and is headquartered in Ahmedabad.
- IN-SPACE Seed Fund Scheme provides grants up to Rs 1 crore to startups and MSMEs.
- Antrix Corporation (1992) is the marketing arm of ISRO.
- FDI: 74% automatic in satellite manufacturing and operations.
- FDI: 49% automatic in launch vehicles and spaceports.
- FDI: 100% in satellite components and subsystems.

Mnemonic - FDI numbers: Seventy-four Satellites, forty-nine Launchers, hundred for Components

74 = 74% automatic FDI in satellite manufacturing and operations; 49 = 49% automatic FDI in launch vehicles and spaceports; 100 = 100% FDI in satellite components and subsystems

Mind-map outline

Policy, Regulators and FDI

- Indian Space Policy 2023
  - Opened value chain to non-government entities
- Institutions
  - IN-SPACE: single-window regulator
  - NewSpace India Limited: ISRO's commercial arm
  - Antrix Corporation: ISRO's marketing arm
- FDI limits by segment
  - 74% automatic: satellite manufacturing, operations
  - 49% automatic: launch vehicles, spaceports
  - 100%: satellite components, subsystems
- Seed Fund Scheme
  - Grants up to Rs 1 crore
  - Startups and MSMEs

## India's first indigenous Hydrogen-powered Train

### Hydrogen train specifications

India's first indigenous Hydrogen-powered Train was inaugurated by PM Modi on the Jind-Sonipat route in Haryana. It is configured as a 10-coach Hydrogen Fuel Cell Trainset powered by a 1,200 kW propulsion system. The train was developed by Integral Coach Factory, Chennai, and its approved operational speed is 75 km/h against a design speed of 110 km/h. Hydrogen is the lightest chemical element, and it makes up nearly 75% of the universe's normal matter.

- Route: Jind-Sonipat, Haryana; length 89 km
- Inaugurated by PM Modi as India's 1st indigenous Hydrogen-powered Train
- Configured as a 10-coach Hydrogen Fuel Cell Trainset
- Propulsion: 1,200 kW Hydrogen Fuel Cell Propulsion System
- Approved operational speed: 75 km/h; design speed: 110 km/h
- Developed by Integral Coach Factory, Chennai
- Hydrogen: lightest chemical element, atomic number 1
- Hydrogen constitutes nearly 75% of the Universe's normal matter

Mind-map outline

Hydrogen train specifications

- Route
  - Jind-Sonipat, Haryana
  - 89 km
- Configuration

- 10-coach trainset
- Propulsion
  - 1,200 kW fuel cell system
- Speed
  - Approved 75 km/h
  - Design 110 km/h

### Components of the hydrogen train

The hydrogen train combines fuel cells, batteries, a hydrogen storage system and traction motors. Proton Exchange Membrane Fuel Cells generate electricity from hydrogen and oxygen. Lithium Iron Phosphate batteries store surplus electricity, while high-pressure cylinders hold the compressed hydrogen needed for propulsion. Electric traction motors convert electrical energy into mechanical energy to drive the train.

- Proton Exchange Membrane Fuel Cells generate electricity through an electrochemical reaction between hydrogen and oxygen
- Lithium Iron Phosphate batteries store surplus electricity generated by fuel cells
- Hydrogen Storage System: high-pressure cylinders store compressed hydrogen required for propulsion
- Electric Traction Motors convert electrical energy into mechanical energy to drive the train
- Four major components: fuel cells, LFP batteries, hydrogen storage, traction motors

Mnemonic - Hydrogen train components: Please Let Cylinders Move

Please = PEM Fuel Cells generate electricity through a reaction of hydrogen and oxygen; Let = LFP batteries store surplus electricity generated by fuel cells; Cylinders = High-pressure cylinders store compressed hydrogen for propulsion; Move = Electric traction motors convert electrical energy into mechanical energy

Mind-map outline

Components of the hydrogen train

- Generate power
  - PEM fuel cells: hydrogen plus oxygen
- Store surplus power
  - LFP batteries
- Store fuel
  - High-pressure cylinders: compressed hydrogen
- Move train
  - Traction motors: electrical to mechanical

### Indian Railways: history and network

Railways were introduced in India in 1853, between Bombay and Thane, during the tenure of Governor General Lord Dalhousie. India ranks fourth globally in railway infrastructure, and the network has 19 zones. Tracks are classified as Broad, Metre or Narrow gauge, with Broad Gauge measuring 1.676 m. The Konkan Railway, constructed in 1998, runs 760 km from Roha to Mangalore.

- First railway line: Bombay to Thane, 1853 (34 km)
- Introduced during Governor General Lord Dalhousie
- India ranks 4th globally in Railway Infrastructure, trailing only US, Russia and China
- 19 Railway Zones in India
- Largest zone: Northern Railway; smallest: North East Frontier Railway
- Newest zone: South Coast Railway
- Gauges: Broad 1.676 m, Metre 1 m, Narrow 0.762 m or 0.610 m
- Mountain railways: Darjeeling Himalayan, Nilgiri Mountain, Kalka-Shimla
- Konkan Railway constructed in 1998
- Konkan Railway: 760 km, Roha (Maharashtra) to Mangalore (Karnataka)

Mnemonic - Track gauge classes: Big Medium Narrow

Big = Broad Gauge, 1.676 m; Medium = Metre Gauge, 1 m; Narrow = Narrow gauge, 0.762 m or 0.610 m

## Mind-map outline

### Indian Railways

- History
  - 1853, Bombay–Thane, 34 km
  - Governor General Lord Dalhousie
- Network
  - 19 railway zones
  - 4th globally in infrastructure
- Gauges
  - Broad 1.676 m
  - Metre 1 m
  - Narrow 0.762 m or 0.610 m
- Konkan Railway
  - Constructed 1998
  - 760 km, Roha to Mangalore

### New developments in railways and S&T

KAVACH is an indigenous Automatic Train Protection System developed by RDSO, with automatic braking and centralised live monitoring. The Vande Metro train has been renamed Namo Bharat Rapid Rail, and Namo Bharat/RapidX runs on the Delhi–Ghaziabad–Meerut corridor. Mission 41K aims to save Rs 41000 crore over 10 years through energy-efficient solutions. The Amrit Bharat Station Scheme plans to redevelop 400 major stations by tying up with private companies.

- KAVACH: indigenously developed Automatic Train Protection System, developed by RDSO
- KAVACH features: centralised live monitoring, automatic braking to prevent over-speeding, relaying SOS messages
- India's 1st Vande Metro Train renamed Namo Bharat Rapid Rail
- Namo Bharat/RapidX: India's first Rapid Rail, Delhi–Ghaziabad–Meerut corridor, operated by NCRTC
- Mission Raftaar: to increase speed of transportation
- Amrit Bharat Station Scheme: redevelopment of 400 major stations with private companies
- Habibganj (near Bhopal) will be India's 1st private railway station
- Mission 41K: save Rs 41000 crore in 10 years through energy-efficient solutions
- Kolkata Metro Extension: India's first underwater metro line, tunnel beneath Hooghly River
- Indian Railways Finance Corporation: established 1986, financial arm of Indian Railways

Mnemonic - KAVACH key features: Live Brakes Save

Live = Centralised live monitoring of train movements; Brakes = Automatic braking for prevention of over-speeding; Save = Relaying SOS messages

## Mind-map outline

### New developments

- Safety
  - KAVACH, developed by RDSO
  - Automatic braking, live monitoring, SOS
- Speed and stations
  - Mission Raftaar
  - Amrit Bharat: 400 major stations
  - Habibganj: first private station
- Energy and metro
  - Mission 41K: Rs 41000 crore savings
  - Kolkata underwater metro
- Trains and finance
  - Namo Bharat Rapid Rail (ex-Vande Metro)
  - Indian Railways Finance Corporation, 1986

## Sports & Awards

### FIFA World Cup 2026

## 2026 Edition: Hosts, Teams and Format

The 23rd men's World Cup was held from 11 June to 19 July 2026 across 16 cities in three countries: the United States, Canada and Mexico. It was only the second World Cup jointly hosted by several nations, after the 2002 edition. The field grew to 48 teams in 12 groups of four, playing 104 matches in total.

- Hosts: United States, Canada and Mexico, the first time three nations jointly hosted
- Dates: 11 June to 19 July 2026; 16 host cities in total
- United States: 11 host cities; Canada: 2 (Toronto, Vancouver); Mexico: 3 (Guadalajara, Mexico City, Monterrey)
- Canada hosted a World Cup for the first time
- Mexico is the first country to host or co-host a World Cup three times (after 1970 and 1986)
- 48 teams in 12 groups of 4; 104 matches in total
- Opening match at Estadio Azteca, Mexico City; final in New Jersey
- Mascots: Maple the Moose (Canada), Zayu the Jaguar (Mexico), Clutch the Bald Eagle (USA)
- Official 2026 song 'Dai Dai' by Shakira and Burna Boy; match ball Adidas Trionda
- Host history: 2002 first held in Asia (South Korea and Japan); 2010 South Africa first African host; 2022 Qatar first Middle East host

Mnemonic - Host city counts: Eleven, Two, Three: United States, Canada, Mexico in that order  
Eleven = United States host cities: 11; Two = Canada host cities: Toronto and Vancouver; Three = Mexico host cities: Guadalajara, Mexico City and Monterrey

Mind-map outline

2026 Edition: Hosts, Teams and Format

- Host nations
  - United States: 11 cities
  - Canada: Toronto, Vancouver
  - Mexico: Guadalajara, Mexico City, Monterrey
- Format
  - 48 teams
  - 12 groups of 4
  - 104 matches
- Symbols
  - Mascots: Maple, Zayu, Clutch
  - Match ball: Adidas Trionda
  - Song: Dai Dai
- Hosting history
  - 2002: South Korea and Japan, first in Asia
  - 2010: South Africa, first in Africa
  - 2022: Qatar, first in Middle East

## Final, Awards and Prize Money

Spain beat Argentina 1-0 in the final at MetLife Stadium, New Jersey, with the winning goal by Ferran Torres in the 106th minute, and England took third place. Individual awards went to Rodri (Golden Ball), Kylian Mbappé (Golden Boot), Unai Simón (Golden Glove) and Pau Cubarsi (Young Player). Team prize money was performance-based and totalled \$671 million across 48 teams.

- Final: Spain 1-0 Argentina at MetLife Stadium, New Jersey
- Winning goal: Ferran Torres, 106th minute
- England secured third place
- Golden Ball (best player): Rodri, Spain captain; Silver Ball: Lionel Messi
- Golden Boot: Kylian Mbappé, 10 goals and 4 assists; his second consecutive Golden Boot
- Silver Boot: Lionel Messi, 8 goals
- Golden Glove (best goalkeeper): Unai Simón
- Young Player Award: Pau Cubarsi
- Fair Play Trophy: Netherlands
- Prize money is performance-based: champion \$50 million; total for all 48 teams \$671 million

Mnemonic - Individual awards: Ready Boys Get Youth

R = Rodri won the Golden Ball; B = Kylian Mbappé won the Golden Boot with 10 goals; G = Unai Simón

won the Golden Glove; Y = Pau Cubarsi won the Young Player Award

Mind-map outline

Final, Awards and Prize Money

- Final
  - Spain 1-0 Argentina
  - Winner: Ferran Torres, 106th minute
  - Venue: MetLife Stadium, New Jersey
- Individual awards
  - Golden Ball: Rodri
  - Golden Boot: Mbappé, 10 goals
  - Golden Glove: Unai Simón
  - Young Player: Pau Cubarsi
- Prize money
  - Performance-based payments
  - Champion: \$50 million
  - Total for 48 teams: \$671 million
- Other awards
  - Third place: England
  - Fair Play Trophy: Netherlands

### **Trophies, FIFA and Title Winners**

FIFA, the global governing body for football, was founded in Paris in 1904 and is headquartered in Zurich. The World Cup has used two trophies: the Jules Rimet Trophy from 1930 to 1970 and the current gold trophy from 1974 onwards. Brazil holds the most titles, with five.

- FIFA: founded in Paris in 1904; headquarters in Zurich, Switzerland
- FIFA has 211 member national associations; President: Gianni Infantino
- Jules Rimet Trophy used 1930 to 1970; made by French sculptor Abel Lafleur
- Brazil won the Jules Rimet Trophy outright in 1970 with a 4-1 win over Italy
- Jules Rimet Trophy was stolen in England before 1966 and found in a rubbish heap by a dog named Pickles
- Current trophy: 18-carat solid gold, designed by Italian sculptor Silvio Gazzaniga
- West Germany were the first to lift the current trophy, with a 2-1 win over Netherlands in 1974
- The current trophy stays with FIFA; winners receive a gold-plated replica
- Titles: Brazil 5; Germany 4; Italy 4; Argentina 3; Uruguay 2; France 2; Spain 2; England 1
- FIFA Peace Prize established in 2025; inaugural prize presented to US President Donald Trump

Mnemonic - Trophy changeover: Rimet retired at Seventy; gold began at Seventy-Four

Seventy = 1970: last tournament using the Jules Rimet Trophy (used 1930 to 1970); Seventy-Four = 1974: first tournament using the current FIFA World Cup Trophy

Mind-map outline

Trophies, FIFA and Title Winners

- FIFA
  - Founded in Paris, 1904
  - Headquarters in Zurich
  - President: Gianni Infantino
- Jules Rimet Trophy
  - Used 1930 to 1970
  - Brazil won it outright in 1970
  - Stolen before 1966, found by dog Pickles
- Current trophy
  - 18-carat gold, used since 1974
  - Designed by Silvio Gazzaniga
  - Winners receive gold-plated replica
- Most titles
  - Brazil: 5
  - Germany and Italy: 4 each
  - Argentina: 3

## Records, Milestones and Firsts

The 2026 tournament produced several individual and team milestones. Lionel Messi became the first man to play at six World Cups, while Cristiano Ronaldo became the first player to score in six consecutive editions. Spain became the first nation to be reigning champion of both the men's and women's World Cups.

- Messi: first man to play at 6 World Cups (2006-2026); eight-time Ballon d'Or winner
- Messi: only player to have won the Golden Ball twice (2014, 2022)
- Ronaldo (Portugal): first player to score in 6 consecutive World Cups (2006-2026)
- Guillermo Ochoa (Mexico): selected for his 6th consecutive World Cup squad
- Spain: first nation to be reigning champion of both the men's and women's World Cups
- Spain: first team to win the World Cup while conceding only one goal in the tournament
- Top 4 ranked teams (Argentina, Spain, France, England) reached the semi-finals for the first time
- Italy, a 4-time champion, failed to qualify for a third consecutive World Cup
- India: only qualification came in 1950, but the team withdrew before the Brazil tournament
- Oldest player: Craig Gordon (Scotland, 43); youngest: Gilberto Mora (Mexico, 17)

Mind-map outline

Records, Milestones and Firsts

- Players
  - Messi: 6 World Cups (2006-2026)
  - Ronaldo: goals in 6 consecutive editions
  - Ochoa: 6th consecutive World Cup squad
- Teams
  - Spain: reigning men's and women's champions
  - Top 4 ranked teams reached semi-finals
  - Italy missed a third straight World Cup
- Debutants and firsts
  - Cape Verde: nicknamed Blue Sharks
  - India: only qualification in 1950
  - 1000th World Cup match: Tunisia v Japan
- Ages
  - Oldest: Craig Gordon, 43
  - Youngest: Gilberto Mora, 17

## Miscellaneous

### Miscellaneous

#### International Relations: Australia, Seychelles and the US

The Prime Minister's visit to Australia for the 3rd India-Australia Annual Leaders' Summit 2026 covered defence, nuclear supply, energy, cyber and sports cooperation. A State Visit to Seychelles, for its 50th National Day, produced digital payment, aid, health and legal agreements. The section also notes the 250th anniversary of US Independence celebrated on 4 July 2026.

- 3rd India-Australia Annual Leaders' Summit held in 2026
- India-Australia bilateral trade reached about USD 50.2 billion in 2025
- Renewed Joint Declaration on Defence and Security Cooperation replaced and expanded the 2009 Framework
- Administrative arrangement under the 2014 Civil Nuclear Agreement enables regular supply of Australian uranium
- Rooftop Solar Training Academy at PDEU Gandhinagar aims to train about 2000 women and youth
- AustralianSuper announced an additional AUD 500 million for India's National Investment and Infrastructure Fund
- Agreement between NPCI International Payments Limited and the Central Bank of Seychelles to facilitate UPI
- India announced a US\$175 million Special Economic Package and a Rupee-based Line of Credit worth 1,250 Crore for Seychelles

Mnemonic - India-Australia summit pillars: DENSE

D = Defence and security cooperation (renewed Joint Declaration); E = Energy security (Joint Statement on Energy Security); N = Nuclear supply of Australian uranium under the Civil Nuclear Agreement; S = Space support for India's Gaganyaan Mission; E = Education (approval of Flinders and Victoria University campuses)

Mind-map outline

International relations

- Australia Leaders' Summit
  - Defence: renewed 2009 framework replaced
  - Nuclear: uranium supply under 2014 deal
  - Solar: Rooftop Solar Academy, Gandhinagar
- Seychelles State Visit
  - Payments: UPI via NPCI International
  - Aid: US\$175 million package
  - Credit: Rupee line of credit 1,250 Crore
- US Independence 250th
  - Celebrated on 4 July 2026

## Cybersecurity and Supercomputing

The Kudankulam nuclear plant faced an alleged ransomware data leak by World Leaks, which renewed concern about the security of critical infrastructure. India's response bodies include CERT-In, I4C, NCIIPC and the National Cyber Forensic Laboratory, while the UN Convention Against Cybercrime is the new global legal instrument. The section also records China's LineShine taking the top spot in the TOP500 supercomputer ranking.

- Kudankulam Nuclear Power Plant in Tamil Nadu is India's largest nuclear power station
- The alleged data leak was claimed by World Leaks, a ransomware group
- Kudankulam reactors are Pressurised Water Reactors developed by NPCIL (India) and Rosatom (Russia)
- CERT-In was established in 2004 in New Delhi as the national nodal agency for cybersecurity incidents
- I4C was established in 2018 under the Ministry of Home Affairs
- NCIIPC was set up in 2014 in New Delhi to protect Critical Information Infrastructure
- National Cyber Forensic Laboratory in Hyderabad gives early-stage forensic help to State/UT Police
- UN Convention Against Cybercrime is the first universal legal instrument; it enters into force 90 days after 40 States ratify
- India is not a signatory to the Budapest Convention on Cybercrime (2001), which took effect in 2004
- China's LineShine tops TOP500 at about 2.19 exaflops, located at the National Supercomputing Center in Shenzhen

Mind-map outline

Cybersecurity and supercomputing

- Kudankulam attack
  - World Leaks ransomware group
  - Largest Indian nuclear power station
  - PWRs by NPCIL and Rosatom
- Cyber bodies
  - CERT-In (2004, New Delhi)
  - I4C (2018, Ministry of Home Affairs)
  - NCIIPC (2014, New Delhi)
- Supercomputing
  - LineShine tops TOP500
  - About 2.19 exaflops, Shenzhen
  - TOP500 published biannually since 1993

## Polity, Law and Minorities

A Joint Parliamentary Committee flagged centralisation and autonomy concerns in the Viksit Bharat Shiksha Adhishthan Bill, which would create one apex regulator for higher education. The Delhi High Court laid down a structured Right to be Forgotten framework that balances privacy against open justice and freedom of expression. The National Commission for Minorities and the constitutional safeguards for minorities complete the section.

- Viksit Bharat Shiksha Adhishthan Bill proposes an apex regulator to replace UGC, AICTE and NCTE
- The proposed Commission works through three councils: Regulatory, Standards and Accreditation
- Regulatory Council prescribes minimum standards for establishing HEIs; Standards Council frames learning outcomes
- Bill excludes funding (grant allocation) from the Commission's mandate
- Appeals are handled by the Central Government, not by an independent mechanism
- Degrees may be awarded only by universities established under Central or State legislation
- Delhi HC in Laksh Vir Singh Yadav vs Union of India (2026) laid down the Right to be Forgotten framework
- Supreme Court in Puttaswamy (2017) recognised the Right to Privacy as a Fundamental Right under Article 21
- NCM is a statutory body constituted on 17 May 1993; members serve a 3-year term; Jains were notified in 2014

Mnemonic - Three councils of the proposed regulator: Regulate, Set, Accredite (RSA)

R = Regulatory Council prescribes minimum standards for establishing HEIs; S = Standards Council frames learning outcomes for HE programmes; A = Accreditation Council develops accreditation norms and frameworks

Mind-map outline

Polity, law and minorities

- VBSA Bill
  - Single apex regulator
  - Three councils: Regulatory, Standards, Accreditation
  - Funding excluded from mandate
  - Appeals with Central Government
- Right to be Forgotten
  - Delhi HC framework, 2026
  - Privacy balanced against public interest
  - Puttaswamy (2017): privacy under Article 21
- Minorities
  - NCM: statutory, constituted 17 May 1993
  - Article 30: minority institutions
  - Article 350B: Special Officer

## Economy and Statistics

The World Bank kept India in the Lower-Middle-Income group for FY2027, using Atlas GNI per capita thresholds that are updated every July. MoSPI's new monthly services production index measures real output of formal services against a base year, and in April 2026 most of its sub-sectors grew at double-digit rates. Inward FDI into India rose sharply in 2025, and Indian firms also raised their overseas investment.

- World Bank FY2027 retained India as a Lower-Middle-Income economy, a status held since 2009
- Classifications use Atlas GNI per capita and are updated every July 1
- Atlas method averages exchange rates over several years to reduce short-term volatility
- FY2027 thresholds: Low up to US\$1,185; Lower-Middle US\$1,186-4,695; Upper-Middle US\$4,696-14,575; High above US\$14,575
- India's GNI per capita in 2025 was USD 2760
- Since 1987, the share of low-income economies fell from nearly 30% to 11%
- New services index covers 19 formal sub-sectors, nearly 60% of the services economy; base year 2024-25
- In April 2026, 14 of 19 service sub-sectors recorded double-digit growth
- India's FDI inflows rose 44% to USD 38.9 billion in 2025, making India the 11th largest FDI recipient

- India's overseas FDI rose 47% to USD 35.7 billion in 2025, making India the 18th largest source of global FDI

Mnemonic - World Bank income groups: Lazy Monkeys Use Hammers

L = Low income: up to US\$1,185 (FY2027); M = Lower-Middle income: US\$1,186 to 4,695; U = Upper-Middle income: US\$4,696 to 14,575; H = High income: above US\$14,575

Mind-map outline

Economy and statistics

- World Bank classification
  - India: Lower-Middle Income since 2009
  - Atlas GNI per capita, updated each July
  - India GNI per capita USD 2760 (2025)
- Services index
  - MoSPI, 19 formal sub-sectors
  - Base year 2024-25
  - 14 of 19 double-digit in April 2026
- Investment
  - Inward FDI USD 38.9 billion (2025)
  - Overseas FDI USD 35.7 billion (2025)
  - India 11th largest FDI recipient

## History, Sports, Awards and Education

The section covers the American Revolution, from the 1765 Stamp Act to the 1783 Treaty of Paris, and the 4 July 1776 Declaration of Independence. It also covers the 2026 Commonwealth Games in Glasgow, the 60th Jnanpith Award and the PGI 2.0 school education grading framework.

- Stamp Act 1765: Britain's first internal tax on American colonists, levied on printed paper
- Boston Tea Party (1773): colonists dumped 342 chests of tea into Boston Harbor
- Battles of Lexington and Concord (1775) marked the beginning of the American Revolutionary War
- Declaration of Independence adopted on 4 July 1776, drafted mainly by Thomas Jefferson
- Battle of Saratoga (1777) helped Benjamin Franklin secure a French military alliance
- Treaty of Paris (1783) formally ended the Revolutionary War
- 23rd Commonwealth Games: Glasgow, 23 July to 2 August 2026, 74 nations and territories, 10 sports
- India's contingent has 125 members including para-athletes; no teams in 3x3 Basketball or Netball
- R. Vairamuthu selected for the 60th Jnanpith Award, the first Tamil poet to receive it
- PGI 2.0 uses grade bands instead of rankings; Chandigarh was the first State/UT to enter Uttam-3 (766/1000)

Mind-map outline

History, sports, awards, education

- American Revolution
  - Stamp Act 1765
  - Boston Tea Party 1773
  - Treaty of Paris 1783
- Commonwealth Games 2026
  - Glasgow, 23 July to 2 August
  - 74 nations and territories
  - India: 125-member contingent
- Awards and education
  - 60th Jnanpith: R. Vairamuthu
  - PGI 2.0: grade bands, not ranks
  - Chandigarh: Uttam-3 (766/1000)

## AILET Corner

### Current Affairs July 2026: AILET Corner

## ICC Women's T20 World Cup 2026

The 10th edition of the biennial ICC Women's T20 tournament was held from 12 June to 5 July 2026, hosted by the England and Wales Cricket Board across 7 venues. The 12 teams played a round-robin group stage in 2 groups of 6, and the top 2 from each group reached the semi-finals. Australia beat England in the final at Lord's to win its record-extending 7th title.

- 10th edition of the biennial ICC Women's T20 International Cricket Tournament
- Held 12 June to 5 July 2026; hosted by England and Wales Cricket Board across 7 venues
- 12 teams played in 2 groups of 6 in a round-robin format
- Top 2 from each group advanced to the semi-finals
- Final at Lord's Cricket Ground, London: Australia defeated England
- Australia won its record-extending 7th title
- Shree Charani (India): leading wicket-taker with 14 wickets
- Charani set a national record for most wickets by an Indian in a single edition
- Beth Mooney (Australia): Player of the Final and Player of the Tournament

Mind-map outline

ICC Women's T20 World Cup 2026

- Format
  - 12 teams, 2 groups of 6
  - Top 2 from each group to semi-finals
- Hosts
  - England and Wales Cricket Board
  - 7 venues
- Result
  - Final at Lord's: Australia beat England
  - Australia's record-extending 7th title
- Awards
  - Charani: 14 wickets, Indian record
  - Beth Mooney: Player of the Final and Tournament

## Sports and Youth Milestones

The Ministry of Youth Affairs and Sports announced that Telangana will host the 8th Khelo India Youth Games in November 2026. Leander Paes was appointed Player Ambassador of the Asian Tennis Federation. Vaibhav Sooryavanshi became India's youngest international cricket debutant at 15 years 99 days.

- Telangana to host the 8th Khelo India Youth Games, scheduled for November 2026 (announced by Mansukh Mandaviya's ministry)
- 7th Khelo India Youth Games were hosted by Bihar in 2025
- Maharashtra was overall champion at the 7th KIYG, followed by Haryana and Rajasthan
- Leander Paes appointed Player Ambassador of the Asian Tennis Federation to promote grassroots tennis
- Paes is India's only tennis player to win an Olympic medal: Bronze in Men's Singles, 1996 Atlanta
- Paes honoured with the Rajiv Gandhi Khel Ratna Award (Major Dhyan Chand Khel Ratna Award)
- Vaibhav Sooryavanshi: India's youngest international cricket debutant at 15 years 99 days
- Sooryavanshi made his senior T20I debut against England at Old Trafford, Manchester
- Sooryavanshi broke Shafali Verma's record (15 years 239 days, against South Africa in 2019)

Mind-map outline

Sports and youth milestones

- Khelo India Youth Games
  - 8th edition: Telangana, November 2026
  - 7th edition: Bihar 2025, Maharashtra champion
- Leander Paes
  - Asian Tennis Federation Player Ambassador
  - 1996 Atlanta Bronze in Men's Singles
  - Rajiv Gandhi Khel Ratna Award
- Vaibhav Sooryavanshi
  - Youngest Indian international debutant

- T20I debut vs England, Old Trafford

## Indian Monsoon

The monsoon is a seasonal reversal of winds that brings major changes in rainfall, and India receives about 75 per cent of its annual rainfall during it. The southwest monsoon runs from June to September and has an Arabian Sea branch and a Bay of Bengal branch. Jet streams influence the monsoon's onset, strength and withdrawal.

- Monsoon: a seasonal reversal of winds bringing significant changes in rainfall; word from Arabic 'Mausim' (season)
- India receives about 75% of its annual rainfall during the monsoon
- Southwest Monsoon: June to September; Northeast Monsoon: October to December
- Arabian Sea branch strikes the Western Ghats and causes heavy rainfall on the west coast
- Bay of Bengal branch moves towards Northeast India and brings rainfall to the Ganga plains
- In summer, land heats faster than water: low pressure over N. India, high pressure over the Indian Ocean
- Himalayas block cold Central Asian winds and force monsoon winds to shed moisture over India
- Jet streams influence the onset, strength and withdrawal of the monsoon
- El Nino (warming of Pacific Ocean waters) can weaken the Indian monsoon and reduce rainfall
- Excess rainfall causes floods; weak monsoons lead to water shortages and droughts

Mnemonic - Three controls on the monsoon: Hills, Jets and El Nino

Hills = Himalayas block cold Central Asian winds and push monsoon winds to shed moisture over India; Jets = Jet streams influence onset, strength and withdrawal of the monsoon; El Nino = Warming of Pacific Ocean waters can weaken the monsoon and reduce rainfall

Mind-map outline

Indian Monsoon

- Concept
  - Seasonal reversal of winds
  - Word from Arabic 'Mausim'
- Branches
  - Arabian Sea branch: Western Ghats
  - Bay of Bengal branch: Ganga plains
- Mechanism
  - Orographic rainfall
  - Himalayas block cold winds
  - Jet streams: onset, strength, withdrawal
  - El Nino weakens monsoon
- Impact
  - Floods from excess rain
  - Droughts from weak monsoon

## Paizhen Fault and the Medog Project

Chinese geologists raised safety concerns over the Medog (Yarlung Tsangpo) Hydropower Project because the active Paizhen Fault lies in the project's reservoir area. The fault is an active tectonic fault in the eastern Himalayan region of south-eastern Tibet and is part of a fault system created by the ongoing collision of the Indian and Eurasian plates. The Yarlung Tsangpo changes its name as it flows from Tibet into India, becoming the Siang and then the Brahmaputra, and finally the Jamuna in Bangladesh.

- Chinese geologists raised safety concerns over the Medog (Yarlung Tsangpo) Hydropower Project
- The concern is linked to the active Paizhen Fault in Tibet
- Paizhen Fault: an active tectonic fault in the eastern Himalayan region of south-eastern Tibet
- The fault passes through the reservoir area of the Medog project
- It forms part of a fault system created by the ongoing collision of the Indian and Eurasian tectonic plates
- Yarlung Tsangpo originates in Tibet
- Enters India through Arunachal Pradesh as the Siang River

- Flows through Assam as the Brahmaputra
- Enters Bangladesh as the Jamuna River

Mnemonic - Yarlung Tsangpo river names: Sam Bought Jam

Sam = Siang River, the name after entering India through Arunachal Pradesh; Bought = Brahmaputra, the name while flowing through Assam; Jam = Jamuna River, the name after entering Bangladesh

Mind-map outline

Paizhen Fault and Medog project

- Concern
  - Chinese geologists' safety worries
  - Active Paizhen Fault
- Fault
  - South-eastern Tibet, eastern Himalayas
  - Indian and Eurasian plate collision
- Yarlung Tsangpo route
  - Originates in Tibet
  - Siang in Arunachal Pradesh
  - Brahmaputra in Assam
  - Jamuna in Bangladesh

## Policy, Heritage and Observances

The Centre released a draft of CAFE-III norms that would tighten fuel-efficiency and CO2 emission standards for passenger vehicles from April 2027. Sushruta, the ancient Indian physician and surgeon, is known as the Father of Surgery, and his bronze statue was unveiled by Scotland's Royal College of Surgeons. July also carries several observances, including the International Day of Cooperatives and World Population Day.

- CAFE stands for Corporate Average Fuel Efficiency
- CAFE norms were introduced by the Bureau of Energy Efficiency in 2017
- Applies to petrol, diesel, CNG, LPG, hybrid and electric passenger vehicles weighing up to 3,500 kg
- Draft CAFE-III norms propose stricter fuel-efficiency and CO2 standards for passenger vehicles from April 2027
- Sushruta: Father of Surgery and Father of Plastic Surgery; disciple of Dhanvantari
- Sushruta authored the Sushruta Samhita; it is part of Ayurveda's Great Trilogy with the Charaka Samhita and Astanga Hridaya
- Sushruta's bronze statue was unveiled by Scotland's Royal College of Surgeons
- International Day of Cooperatives: 1st Saturday of July; 2026 theme 'Cooperatives for a Peaceful World'
- National Statistics Day: 29 June, honouring Prasanta Chandra Mahalanobis; 2026 theme 'Unlocking the Potential of Administrative Data'
- World Population Day: 11 July; 2026 theme 'Realizing the Hopes and Aspirations of Young People - Today and for the Future'

Mind-map outline

Policy, heritage and observances

- CAFE-III norms
  - Draft released by the Centre
  - Stricter fuel efficiency and CO2 standards
  - Applicable from April 2027
- Sushruta
  - Father of Surgery
  - Authored Sushruta Samhita
  - Part of Ayurveda's Great Trilogy
- Observances
  - Cooperatives Day: 1st Saturday of July
  - Statistics Day: 29 June
  - World Population Day: 11 July